Installation and Maintenance Instructions

Screw Vacuum Pumps

COBRA NC 630 B

Cooling water system with plate heat exchanger

Ateliers Busch S.A.
Zone industrielle
2906 Chevenez
Switzerland

0870556803 / 211211 / Original instructions / Modifications reserved

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Introduction

Congratulations on your purchase of the Busch vacuum pump. With watchful observation of the field’s requirements, innovation and steady development Busch delivers modern vacuum and pressure solutions worldwide.

These operating instructions contain information for

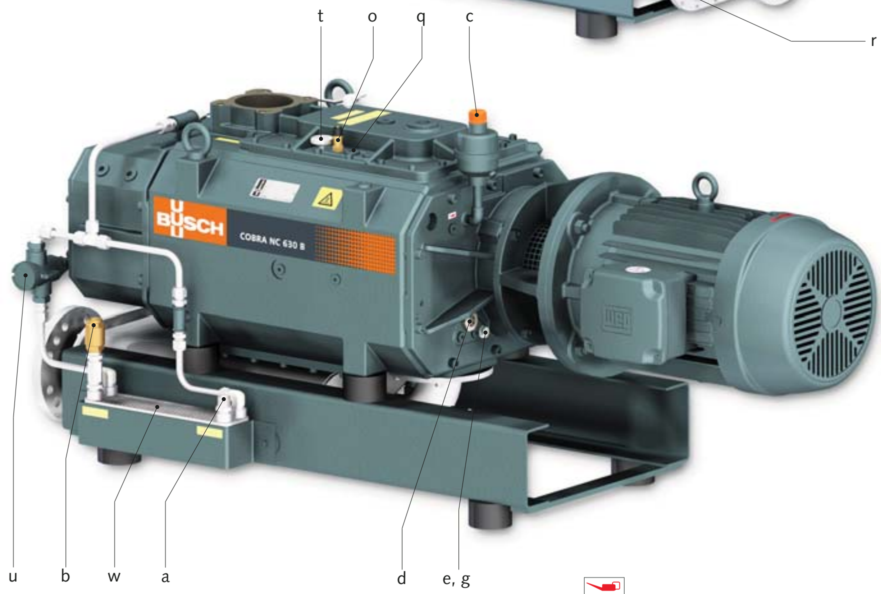
- product description,
- security,
- transport,
- storage,
- installation and commissioning,
- maintenance,
- overhaul,
- trouble shooting

of the the vacuum pump.

For the purpose of these instructions, “handling” the vacuum system means the transport, storage, installation, commissioning, influence on operating conditions, maintenance, troubleshooting and overhaul of the vacuum system.

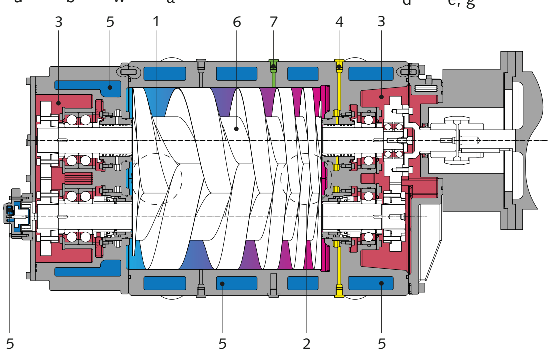
Prior to handling the vacuum system, these operating instructions shall be read and understood. If anything remains to be clarified please contact your Busch representative!

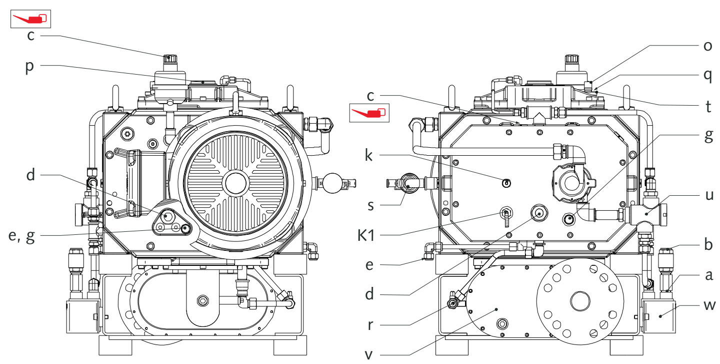
Keep these operating instructions and, if applicable, other pertinent operating instructions available on site.



a Cooling water inlet
b Cooling water outlet
c Oil filler plug (with filter for mechanical seal version)
d Oil sight glass
e Oil drain plug
g Magnetic plug
k Plug for manual rotation of rotors
o Safety valve
p Cooling liquid filler plug
q Cooling liquid vent plug
r Cooling liquid purge ball valve
s Gas ballast (option)
t Thermometer
u Temperature regulator
v Silencer (accessory)
w Plate heat exchanger
K1 Temperature switch
K1 Resistance thermometer PT 100 (option)

1 Inlet
2 Discharge
3 Oil
4 Barrier gas (option)
5 Cooling liquid
6 Screw rotors
7 Dilution gas (option)





Product Description

Use

The COBRA NC vacuum pumps are designed for use in the field of industrial applications and similar industries.

They can be used to draw gases and gas mixtures.

> ⚠ WARNING
>
> **When using toxic, inflammable and/or explosive gases, make sure that the system corresponds in design to applicable local and national safety regulations and that all applicable safety measures are followed.**
> **All product-specific safety regulations must be observed.**

Solid particles must not get into the vacuum pump. Procedural errors can result in the pump drawing in a certain quantity of liquid. If the pump has drawn in liquid, a short drying time will be necessary at the end of the process.In case a silencer (v) (accessory) is fitted at the outlet:

- Drain the silencer (v) (accessory)

The maximum admissible temperature of drawn gas depends on the inlet pressure and the nature of the gases. The lower the inlet pressure (Pa), the higher can be the drawn gas temperature (TGas). We can consider as air the following indications:

– Pa < 50 mbar, TGas < 200°C
– Pa > 50 mbar, TGas < 70°C

The vacuum pump is designed for use in an environment which is non-explosive.

The vacuum pump is thermally suitable for continuous operation at any pressure between atmospheric pressure and ultimate pressure.

Principle of operation

The COBRA NC vacuum pumps are screw vacuum pumps with cooling water circuit.

The COBRA NC vacuum pumps operates according to the principle of screw pumps. Two parallel screws (6) rotate in opposite directions in the pump body. Entering gases are trapped between the pitches of the screws and the pump body. The gases are conveyed by the rotation of the screws to the exhaust side where they are discharged.

The COBRA NC vacuum pump is driven by an air-cooled motor.

Cooling

The vacuum pump is cooled by a cooling liquid (mix of water and glycol) inside the water chambers (5) of the cylinder and the inlet side end plate (B- side). This indirect circuit is done using an integrated recirculating pump.

The cooling liquid is cooled by a plate heat exchanger (w) which must be connected up to the water main.

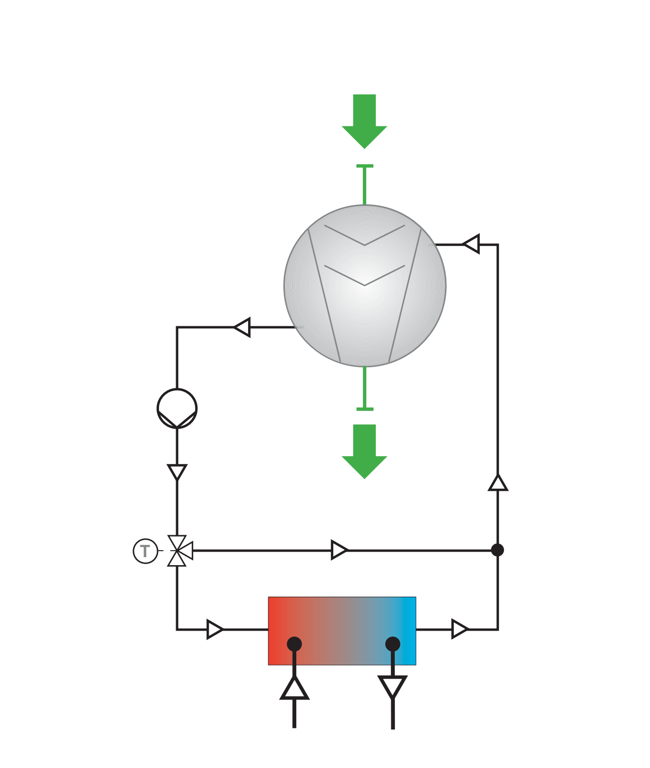
The cooling circuit is equipped with a temperature regulator (u) fitted upstream of the plate heat exchanger (w) .When the cooling liquid temperature exceeds 55 °C, the temperature regulator opens (mechanical opening) and allows the cooling liquid to get into the heat exchanger.

The temperatures switch (K1) enables the monitoring of the oil temperature in the cylinder endplate. It sends a warning signal when the oil temperature exceeds 85 °C. 30 seconds after the warning signal goes off, the pump must stop.

The thermometer (t) allows a visual display of the cooling liquid temperature inside the cylinder of the vacuum pump.

NOTE:
The COBRA NC vacuum pumps are shipped without oil, cooling water and cooling liquid. Operation without any of these lubricants will destroy the vacuum pump within a short period of time.

Process diagram



Sealing systems

The COBRA NC vacuum pumps are equipped with labyrinth seals on the motor side (A-side) and inlet side (B-side) as standard.

Operational Options / Use of Available Accessories

The relief valve (o) prevents excessively high pressure in the cylinder, relief pressure: 6 bar.

A silencer or sound absorber (v) (accessory) at the outlet of the pump reduces the noise of the pump and collects any condensates.

A resistance thermometer PT 100 (K1) (option) allows the monitoring of the oil temperature. When the temperature exceeds 85°C, the pump must be switched off.

A gas ballast for ambient air (s) (option). This device is fitted with a manual isolation valve which can be opened when process conditions require its use.

The gas ballast increases the pumps capacity to handle condensable vapours that may be present in the process stream.

The gas ballast has an influence on the ultimate pressure of the pump (see table below).

Operating frequency	Ultimate pressure		
	standard	with closed gas ballast	with opened gas ballast for ambient air
50 Hz	0.05 hPa	0.05 hPa	0.5 hPa
60 Hz	0.01 hPa	0.01 hPa	0.1 hPa

A nitrogen supply system fitted to the base frame allows the supply of nitrogen to a number of different points on the vacuum pump. The system allows to adjust pressure and volume flow separately. This gas can be used in a number of different ways:

- A purge gas system fitted to the inlet flange allows to flush the vacuum pump after use or during operation. This system consists of a solenoid valve which enables to open and close the purge circuit. The filtered gas is fed directly into the inlet flange.
- Barrier gas for the labyrinth seals or the oil-lubricated single mechanical seal: this option seals off the process gases and the gear oil. The nitrogen is fed into the intermediary chambers (4).
- Dilution gas: this option prevents the formation of condensates or dilutes them, depending on the application. The nitrogen is fed directly into the cylinder (7) of the vacuum pump.

On / Off switch

The vacuum pump is delivered without on / off switch. The control of the vacuum pump must be provided in the course of the installation.

Safety

Intended use

DEFINITION: For the purpose of correct understanding, the “handling” of the vacuum pump implies the transport, storage, installation, commissioning, the influence on operating conditions, maintenance, troubleshooting and overhaul of the vacuum pump.

The vacuum pump is intended for industrial use. It should only be handled by qualified staff.

The different applications for use and operational limits of the vacuum pump as laid out in the “Product Description” and the “Installation Prerequisites” of the vacuum pump must be observed both by the manufacturer of the machinery into which the vacuum pump is to be incorporated and by the end user.

The need for personal safety regulations depends mainly on the application the pump will be used in. The end user must provide the operators with all necessary means and tools and must inform his personnel about any dangers emanating from the processed products.

The operator of the vacuum pump must observe the safety regulations and must train and instruct his personnel accordingly.

Local regulations regarding the motors and electric control elements must be observed when installing the pump in potentially explosive environments.

The maintenance instructions must be observed and respected.

Is is vital that these installation and maintenance instructions are read and understood before the vacuum pump is used. If you have any doubts, please contact your local Busch representative.

Safety information

The vacuum pump has been designed and manufactured in accordance with the latest technical and safety standards. Nevertheless, residual risks may remain.

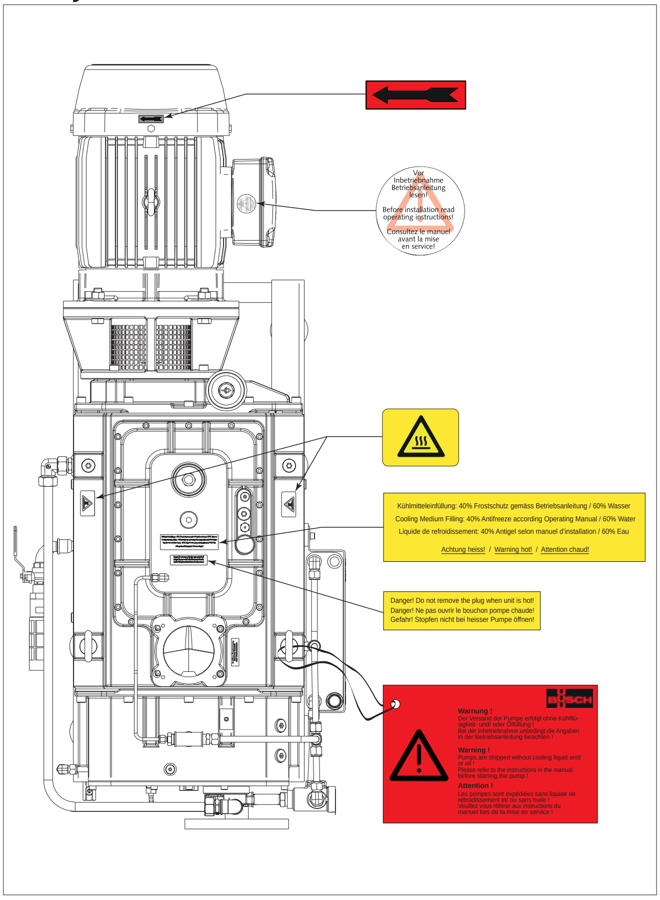
A lot of safety information is mentioned in these Installation and Operating Instructions as well as on the pump. The safety instructions must be observed. The safety information can quickly be detected through key words like DANGER, WARNING and CAUTION and is defined as follows:

> DANGER
>
> **Disregard of this safety note will always lead to accidents with potentially fatal injuries and serious damages.**

> WARNING
>
> **Disregard of this safety note may lead to accidents with potentially fatal injuries and serious damages.**

> CAUTION
>
> Disregard of this safety note will always lead to accidents with minor injuries and damages to property.

Safety stickers



Noise emission

For the sound pressure level in free field according to EN ISO 2151, see "Technical Data".

> CAUTION
>
> The vacuum pump emits noise of high intensity.
>
> Risk of damage to the hearing.
>
> Persons staying in the vicinity of a non noise insulated vacuum pump over extended periods shall wear ear protection.

Safety area

Before any maintenance action, ensure a safety perimeter of at least 1 [m] around the pump.

Stopping procedure for maintenance

- Stop the vacuum pump
- Switch off power supply (the vacuum pump must be fully disconnected from any power supply)
- Disconnect water connections (inlet first, then outlet)
- Put up label or warning board "Maintenance in progress" on or next to the pump.

Start-up procedure after maintenance

- Remove label or warning board "Maintenance in progress"
- Check cooling liquid and oil levels according to the chapters "Checking the oil level" and "Checking the cooling liquid level"
- Switch on power supply (the vacuum pump must be connected to the power supply)
- Connect the water connections (outlet first, then inlet)
- Check that the "Installation prerequisites" are observed
- Start the vacuum pump

Transport

The COBRA NC vacuum pumps are tested and checked in our factory before careful packing. Check the packaging for transport damage when the goods arrive. The pump can withstand temperatures betweeen -25°C and +55°C during transport.

Transportation of packaged pump

Packed on a pallet, the vacuum pump can be transported with a forklift.

Transport in unpacked state

The vacuum pump is bolted to a pallet or a base plate:

- Remove the bolts between the vacuum pump and the pallet or base plate

> CAUTION
>
> Do not walk, work or stand under suspended loads.

> CAUTION
>
> Please check out the weight of the vacuum pump before lifting it up (see "Technical Data").
>
> Use adequate lifting gear for this.

NOTE: The eyebolts are fitted more or less at equal distance from the centre of gravity of the vacuum pump incl. drive motor. If any accessories that could upset the balance of the vacuum pump, are installed, or if the vacuum pump is delivered without drive motor, it is necessary to add a belt or rope at a suitable point when lifting the pump.

- Attach lifting gear securely to the eyebolt or eyebolts
- Use lifting gear with a crane hook equipped with safety latch.
- Lift the vacuum pump

In case the vacuum pump was bolted to a pallet with fixing bolts:

- ◆ Unscrew the fixing bolts in the base frame

CAUTION

Once filled with oil, the vacuum pump cannot be lifted anymore.

Prior to every transport make sure that the oil has been drained from the vacuum pump

The packaging material must be disposed of in accordance with local and national regulations.

Storage

Short-term Storage

- Make sure that the inlet / discharge connections are closed
- Store the vacuum pump

– if possible in its original packaging,
– indoors,
– dry,
– in a dust free room and
– free from vibrations

Removal from storage

Before starting up a vacuum pump that has been stored outside the building for a while, the vacuum pump must be moved to a room with ambient temperature, where it should rest for a day.

Conservation

In case of adverse ambient conditions (e.g. aggressive atmosphere, frequent temperature changes) conserve the vacuum pump immediately. In case of favourable ambient conditions conserve the vacuum pump if a storage of more than 3 months is anticipated.

- Make sure that all openings are hermetically sealed; use adhesive tape to fasten loose parts (such as o- rings, flat seals, etc.).

NOTE: VCI is the abbreviation for “volatile corrosion inhibitor”. The VCI molecule is an organic corrosion inhibitor in the vapour phase. Integrated in various carriers such as film, cardboard, paper, foam, liquid and powder, it protects parts against corrosion as a result of its action in vapour phase. However, VCI packaging can attack plastic surfaces and surfaces of other elastomers. If in doubt, please contact your nearest distributor. VCI packaging provides several years of protection against corrosion, even under harshest conditions: overseas shipment, extended storage before use.

- Wrap the vacuum pump in VCI film
- Store the vacuum pump

– if possible in its original packaging,
– indoors,
– dry,
– in a dust free and
– vibration free area

Starting-up of the vacuum pump after storage

- Please ensure that all protective agents such as gaskets, plugs or adhesive tapes that were used for the protection of the pump, are removed.
- Commission the vacuum pump as described in chapter “Installation and Commissioning”

Installation and Commissioning

Installation prerequisites

CAUTION

In case of non-compliance with the installation prerequisites, particularly in case of insufficient cooling:

Risk of damage or destruction of the vacuum pump and its components!

Risk of personal injury!

The installation prerequisites must be complied with.

- Please ensure that the integration of the vacuum pump is compliant with the safety requirements of the Machine Directive 2006/42/EC (concerning the responsibility of the system’s manufacturer into which the vacuum pump is to be incorporated, please also refer to the note in the EC-Declaration of Conformity).

WARNING

Local regulations regarding the motors and electric control elements must be observed when installing the pump in potentially explosive environments. Before start- up, make sure that all safety measures have been followed.

Local installation

- Make sure that the environment of the vacuum pump is not potentially explosive
- Make sure that the following ambient conditions are adhered to:

– Ambient temperature: 5 ... 50 °C
– Ambient pressure: atmospheric
– Humidity range: 20 à 95%
– Altitude: up to 1000 m

- Make sure that the cooling water fulfills the following requirements:

– Temperature: 10 - 30 °C
– Water pressure: 1 - 6 bar (relative)
– Water flow rate: 12 l/min
– Water pressure differencial between cooling water inlet and outlet: ≥ 1 bar
– Water hardness: < 5° dGH

NOTE: 1° (german degree = 1° dGH) = 1,78° (french degree) = 1,25 e (english degree) = 17,9 mg/kg CaCO3 (american hardness)

- Make sure that the cooling water is neutral and clean
- Make sure that the environment conditions correspond to the protection class of the motor (according to motor nameplate)
- Make sure that the vacuum pump is placed on or fastened to a horizontal surface
- Make sure that the vacuum pump is level and even
- Make sure that the vacuum pump cannot inadvertently or intentionally be used as a support for heavy objects
- Make sure that the vacuum pump cannot be hit by falling objects
- Make sure that the vacuum pump is at least 1 m away from any wall
- Make sure that the vacuum pump is easily accessible and that the selected installation site fulfills the requirements for assembly / dismantling

- Make sure that no temperature-sensitive part (such as plastic, wood, cardboard, paper, electronic parts) come into direct contact with the hot surfaces of the vacuum pump
- Make sure that the installation site or assembly area is ventilated in such a way that adequate cooling of the vacuum pump is assured

CAUTION

During operation the surface of the vacuum pump may exceed temperatures of 90° C.

Risk of burns!

- Make sure that the vacuum pump cannot be touched inadvertently during operation, provide a guard if necessary
- Make sure that the oil sight glasses (d) will remain easily accessible

If the oil change is meant to be performed on site:

- ◆ Make sure that the oil drain plugs as well as the oil filler plugs, are easily accessible.

Suction Connection

- Make sure that the protection that prevents the ingress of foreign matter during transport, has been removed before connecting up the vacuum pump to the piping.

CAUTION

Do not put hands into the inlet aperture.

Risk of body damage !

CAUTION

The ingress of foreign objects or liquids can destroy the vacuum pump.

In case the inlet gases can contain dust or other foreign solid particles:

- ◆ Make sure that a suitable filter or protection screen is installed at the inlet of the vacuum pump

- Make sure that the nominal diameter of the inlet line is at least equal to the diameter of the inlet flange to prevent a drop in the performance of the vacuum pump in the case of a smaller cross-section.
- Make sure that the vacuum pump is connected with leak proof lines.

CAUTION

Once the inlet line has been connected up, make sure that the system does not leak. Leakage of dangerous substances must be prevented!

- Make sure that the inlet line is equipped with a shut-off device upstream of the inlet flange, so that the flow of drawn gases can be stopped
- Make sure that the inlet line does not exercise any load on the inlet flange. Use bellows if necessary.
- The inlet flange has the following dimension:

– DN 100 ISO-K

In case of long inlet lines the pipe diameter should be larger than the inlet flange to prevent a drop in the performance of the vacuum pump. If you have any doubts, contact your Busch representative.

Discharge connection

CAUTION

Do not put hands into the outlet aperture.

Risk of body damage !

The following instructions for connection to the discharge side only apply if the drawn gas is discharged into a suitable environment by the vacuum pump.

- Make sure that the protection, that was fitted to prevent the ingress of particles during transport, has been removed before the vacuum pump is connected up to the vacuum pipe
- Make sure that the nominal diameter of the discharge line corresponds at least to the diameter of the exhaust flange of the vacuum pump in order to prevent a drop in the performance of the vacuum pump, in case of use of a smaller cross-section

CAUTION

When the discharge piping has been connected up, make sure that the system does not leak. Leakage of dangerous substances must be prevented!

- Make sure that the discharge line is fitted in such a way so as to prevent any liquids or condensates to re-enter the vacuum pump (siphon, discharge line sloping away from the pump)
- Make sure that the discharge line is not equipped with a shut-off
- Make sure that the discharge line does not exercise any load on the exhaust flange. Use bellows if necessary
- The outlet flange has the following dimension:
  - ◆ Pump outlet
    – DN 100 ISO-K, DIN 28404
  - ◆ Silencer outlet (v) (accessory)
    – DN 80 PN16, EN1092-1
    – ANSI 3" 150 lbs
- Maximal counter pressure at the discharge:

– 0,2 bar

In the case of long discharge lines, the line cross-section should be larger than the exhaust flange in order to prevent a drop in the performance of the vacuum pump. If you have any doubts, contact your Busch representative.

Cooling water connections

The cooling water is generally connected up with a flexible hose.

The cooling water outlet must be unpressurised.

Connection diameter:

– cooling water inlet: G 1/2
– cooling water outlet: G 1/2

Nitrogen system connections

The connection of the nitrogen system is generally done using flexible hoses (diameter 1/4")

Connection diameter:

– G 1/4, ISO 228-1

Electrical connection / Checks

- Make sure that the requirements according to EMC-Directive 2004/108/EC and Low-Voltage-Directive 2006/95/EC as well as the current EN-standards, electrical and occupational safety directives and the local or national regulations respectively, are complied with (this is the responsibility of the designer of the machinery into which the vacuum pump is to be incorporated; see also the corresponding comments in the EC-Declaration of Conformity).

- Make sure that the power supply is compatible with the data on the nameplate of the drive motor
- Make sure that an overload protection according to EN 60204-1 is provided for the drive motor
- Make sure that the drive of the vacuum pump will not be affected by electric or electromagnetic interference; if unsure please seek advice from your Busch representative

Temperature control

The oil temperature in the gearbox is monitored by a temperature switch (K1) that is mounted in the cylinder endplate.

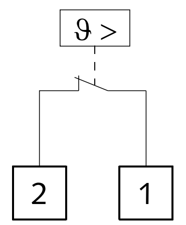
The electrical wiring of the temperature switch has to be done in order that the vacuum pump stops when the temperature switch trips (break contact).

Designation		Set value [°C]
K1	Cylinder endplate	85

Technical data

Technical data	
Set value	85 °C ±5 °C
Switching cycles	10'000
AC current cosφ = 1	2,5 A / 250 VAC
AC current cosφ = 0,6	1,6 A / 250 VAC
DC current	1,6 A / 24 VDC
DC current	1,25 A / 48 VDC
Resistance contact max.	50 mΩ
Bounce time contact max.	1 ms
High voltage max.	2 kV
Cable operating temperature max.	motionless -30°C ... +80°C
Cable operating temperature max.	moved -5°C ... +70°C

Electrical wiring



Resistance thermometer PT 100 (option)

Resistance thermometer PT 100 instead of temperature switch.

Monitoring the barrier gas volume flow

A flow meter with contact fitted to the control panel of the nitrogen system enables the monitoring of the gas volume flow which is fed into the intermediary chambers (4).

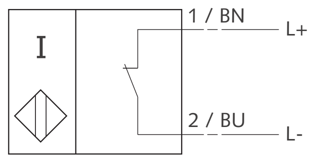
The flow meter must be connected in such a way that the vacuum pump switches off when the volume flow drops below the min. set flow value.

– Set value: 3,5 Nl/ min

Technical data

Technical data	Flow meter
assigned working voltage	16 V
Current consumption	52 mA
Control range	0,75 - 5,75 Nl/ min
Pressure	3 bar

Electrical wiring



Solenoid valve

Opening/ closing of the nitrogen barrier gas circuit

A solenoid valve connected to the barrier gas circuit enables to control the opening/ closing of the nitrogen barrier gas circuit.

The nitrogen barrier gas circuit depends on the application and is defined by the end user of the vacuum pump.

Opening/closing the nitrogen purge gas circuit

A solenoid valve, connected to the inlet flange, enables to control the opening / closing of the nitrogen purge gas circuit.

The nitrogen purge gas circuit depends on the application and is defined by the end user of the vacuum pump.

Opening/ closing the nitrogen dilution gas circuit

A solenoid valve connected to the dilution gas circuit enables to control the opening/ closing of the nitrogen dilution gas circuit.

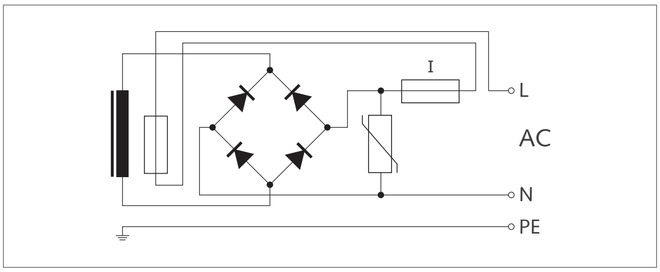
The dilution gas circuit depends on the application and is defined by the end user of the vacuum pump.

Technical data

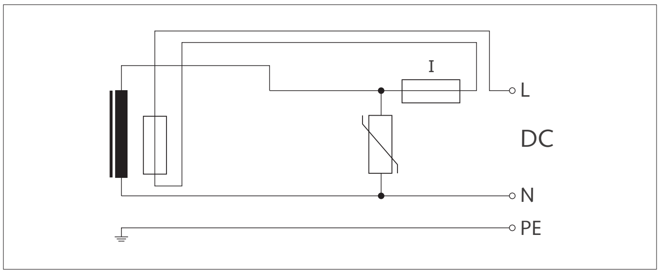
Technical data	Solenoid valve
Nominal AC voltage	12 - 380 VAC
Nominal DC voltage	12 - 220 VDC
Ambient temperature	-40 ... +75 °C
Current AC consumption	Pn (hot), 9 W P (cold) 20°C, 11 W
Current DC consumption	Pn (Maintenance), 11 W cold attraction 13 W

Electrical wiring

Electrical connection with alternating current



Electrical connection with direct current



Installation

Fitting

- Make sure that the "Installation Prerequisites" are complied with
- Fit or install the vacuum pump at its final location

Electrical connection

WARNING

Risk of electrical shock, risk of damage to equipment.

Electrical installation work must only be executed by qualified personnel that knows and observes the following regulations:
- IEC 364 or CENELEC HD 384 or DIN VDE 0100, respectively,
- IEC-Report 664 or DIN VDE 0110,
- BGV A2 (VBG 4) or corresponding national accident prevention regulation.

CAUTION

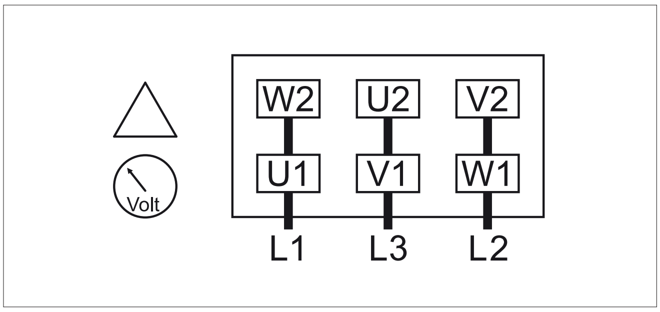
The connection schemes given below are typical. Specific orders or deviating connection schemes for certain markets may apply.

Risk of damage to the drive motor!

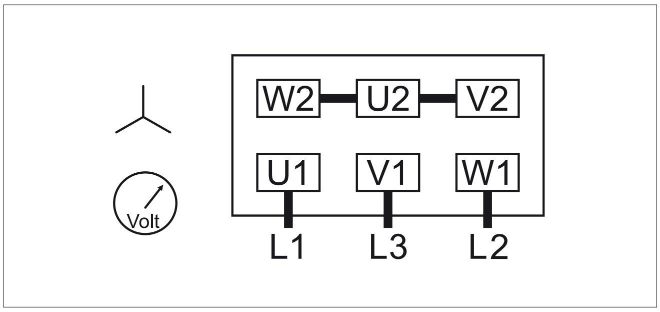
Please check drive motor connections inside the terminal box and refer to the drive motor connection instructions.

- Connect the drive motor electrically
- Connect the earth line

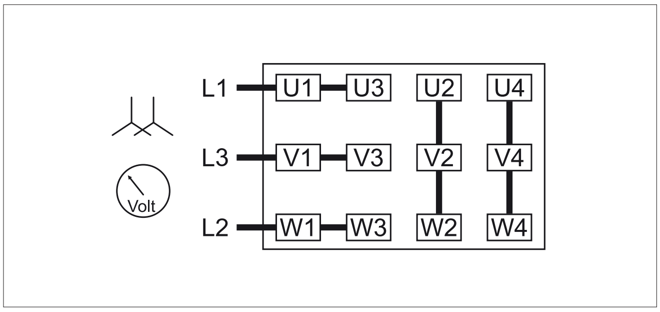
Delta connection (Low voltage):



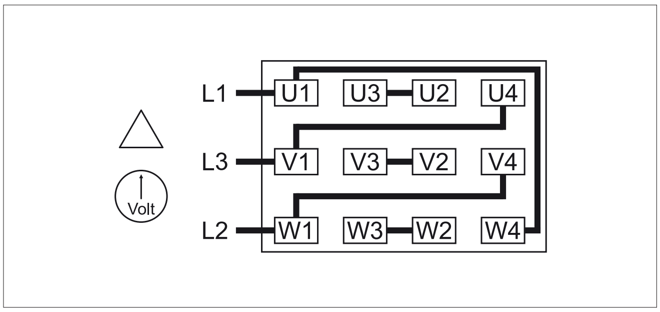
Star connection (High voltage):



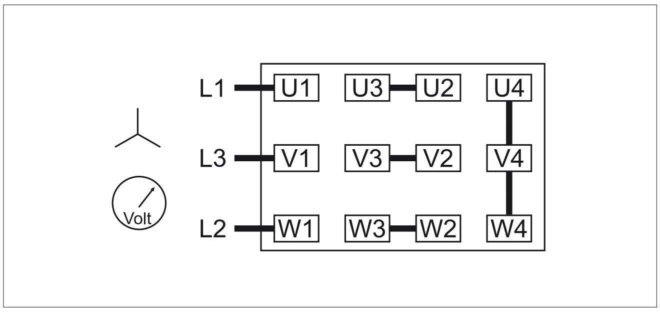
Double star connection, multi-voltage motor with 12 terminals (Low voltage):



Delta connection, multi-voltage motor with 12 terminals (Middle voltage):



Star connection, multi-voltage motor with 12 terminals (High voltage):



CAUTION

Operation of the pump in the wrong direction of rotation, even for a short period of time, can destroy the vacuum pump.

Risk of damage of the drive motor!

Prior to starting-up of the vacuum pump, please ensure that the vacuum pump is connected up correctly.

- Determine the direction of rotation with the arrow sign on the motor (see image "Safety stickers").
- Press the start button and hold briefly
- Watch the fan of the drive motor and determine the direction of rotation just before the fan stops

If the fan is turning in the wrong direction:

- ◆ Switch around any two of the drive motor wires in the terminal box

Connecting up of lines/ pipes

- Connect the suction line
- Connect the discharge line
- Make sure that all provided covers, guards, hoods etc. are fitted

Filling up with oil

CAUTION

It is possible that the vacuum pump was tested with an oil type different that that will be used for your application. The vacuum pumps tested with a special oil are dispatched with specific stickers (“Special Oil”) on the gear box housing like on the cylinder cover inlet side (B-side).

If the oil type is not compatible, want to carry out a cleaning of the parts in contact with oil.

Make sure that the bearings are lubricated before the reassembly.

The COBRA NC vacuum pumps are delivered without oil (see chapter “Oil Type” for information on recommended oils).

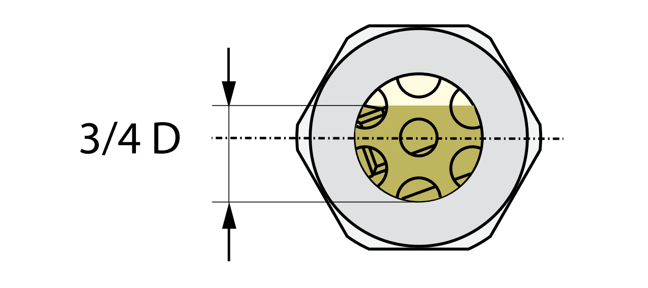
- Prepare the quantity of oil specified in the table “Oil quantity”

NOTE: The quantity of oil specified in the operating instructions is of informative nature only. Check the oil level with the help of the various oil sight glasses (d) on the vacuum pump.

CAUTION

Before changing the type of oil, make sure that the new type is compatible with the old type. If necessary, flush the vacuum pump.

- Unscrew the oil filler plugs (c)
- Fill in oil
- Make sure that the oil level still is in the top quarter of the oil sight glasses (d)



- Make sure that the seal ring in the oil filler plugs is not damaged, replace plugs if necessary
- Fit the oil filler plugs and tighten up

NOTE: Starting the vacuum pump when the oil is cold is made easier when the suction line is left open or the rubber mat is removed.

- Switch on the vacuum pump

In case the suction line is equipped with a shut-off valve:

- ◆ Close the shut-off valve

In case the suction line is not equipped with a shut-off valve:

- ◆ Cover the inlet connection with a rubber mat

- Let the vacuum pump run for a few minutes
- Switch of the vacuum pump and wait for a few minutes
- Make sure that the oil level still is in the top quarter of the oil sight glasses (d)

If the oil level is below the middle of the oil sight glasses:

- ◆ Top-up with oil

In case the suction line is equipped with a shut-off valve:

- ◆ Open the shut-off valve

In case the suction line is not equipped with a shut-off valve:

- ◆ Remove the rubber mat from the suction flange and connect the suction line to the suction flange

- Before any transport, make sure that the oil has been drained out of the vacuum pump.

CAUTION

Once filled with oil, the vacuum pump should not be lifted or moved anymore.

CAUTION

The vacuum pump must remain in a horizontal position once it has been filled with oil.

Filling in cooling liquid

The COBRA NC vacuum pumps are delivered without cooling liquid (see chapter “Cooling liquid types ” for information on the recommended cooling liquids).

- Prepare the quantity of cooling liquid as specified in the table “Cooling liquid quantity”

NOTE: The quantity of cooling liquid specified in the installation handbook is of informative nature only. Follow the procedure of filling the cooling liquid.

- Remove the cooling liquid filler plug (p) on cylinder upper plate
- Open the cooling liquid drain plug (q1) on cylinder upper plate
- Close the cooling liquid vent plug (q) on cylinder upper plate
- Fit the cooling liquid filler plug (p) on cylinder upper plate
- If there is any spillage on the surface of the vacuum pump, wipe it up
- Start the vacuum pump

If the inlet line is equipped with a shut-off device:

- ◆ Close the shut-off device

If the inlet line is not equipped with a shut-off device:

- ◆ Place a rubber mat on the suction flange

- Let the vacuum pump run for maximum 5 minutes
- Stop the vacuum pump and wait for a few minutes
- Open the cooling liquid vent plug (q) on cylinder upper plate
- Check that the filling level is just under the cylinder upper plate

In case the cooling liquid level is below the required level:

- ◆ Fill in more cooling liquid

If the inlet line is equipped with a shut-off device:

- ◆ Open the shut-off device

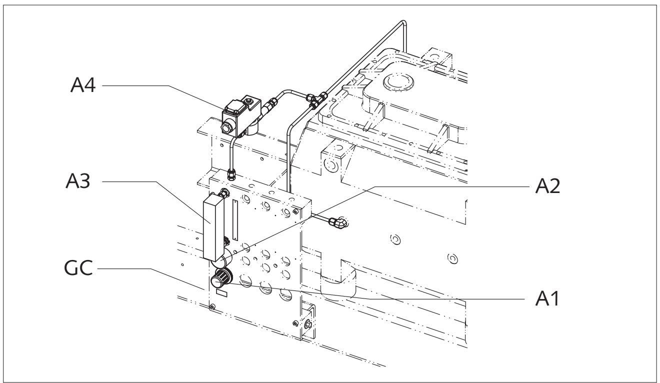
If the inlet line is not equipped with a shut-off device:

- ◆ Remove the rubber mat from the suction flange and connect the suction line to the suction flange

Nitrogen system

Barrier gas

Connection/ control panel



GC	Gas connection G 1/4
A1	Pressure regulator
A2	Manometer
A3	Flow meter with contact
A4	Solenoid valve

Set values

Type	Volume flow [Nl/ min]	Over-pressure [bar]
NC 630 B	4 ... 5	3

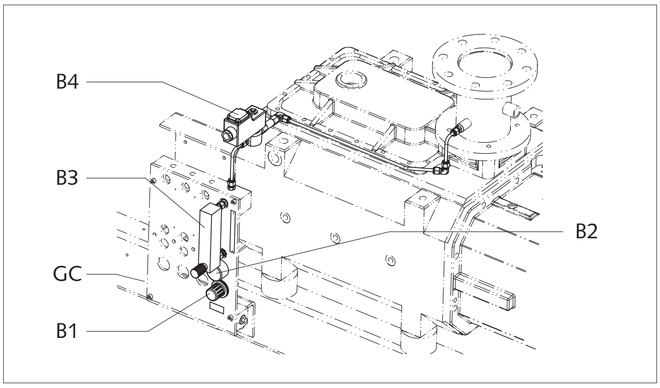
Adjustment of the barrier gas:

- Connect the different supervisory instruments (flow meter with contact, solenoid valve)
- Connect the nitrogen main supply up to the connection on the control panel (GC)
  - ◆ Over-pressure: 3 bar
  - ◆ Gas type: nitrogen or compressed air
- Adjust the barrier gas pressure with the help of the pressure regulator (A1) to 3 bar (relative)
  - ◆ The gauge (A2) shows a visual display of the barrier gas pressure

NOTE: The barrier gas volume flow is approx. 4 Nl/min. The barrier gas volume flow should not drop below the set flow value.

Purge gas

Connection/ control panel



GC	Gas connection G 1/4
B1	Pressure regulator
B2	Manometer
B3	Flow meter with regulating valve
B4	Solenoid valve

Set values

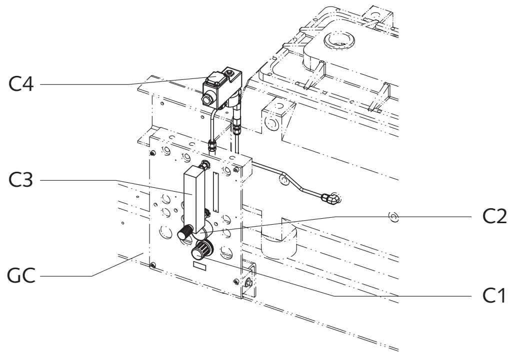
Type	Volume flow [Nl/ min]	Over-pressure [bar]
NC 630 B	≥ 100	2,5

Adjustment of the purge gas:

- Connect the solenoid valve (B4)
- Connect the nitrogen main supply up to the connection on the control panel (GC)
- Switch on the COBRA NC vacuum pump, the inlet is closed
  - ◆ Open the solenoid valve (B4) of the purge gas circuit
- Adjust the purge gas pressure with the help of the pressure regulator (B1) to 2,5 bar (relative)
  - ◆ The gauge (B2) shows a visual display of the barrier gas pressure
- Adjust the purge gas volume flow with the help of the regulating valve of the flow meter (B3)
  - ◆ The purge gas volume flow must correspond to the set flow value
- Close the solenoid valve (B4) of the purge gas circuit

Dilution gas

Connection/ control panel



GC	Gas connection G 1/4
C1	Pressure regulator
C2	Manometer
C3	Flow meter with regulating valve
C4	Solenoid valve

Set values

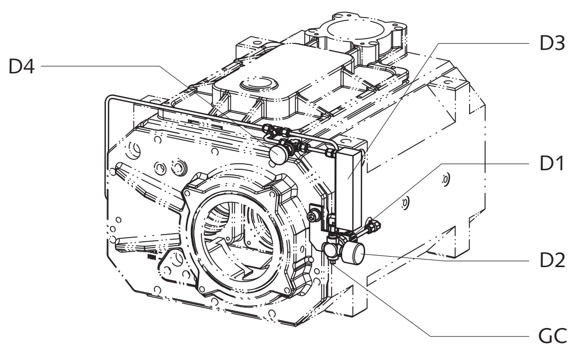
Type	Volume flow [Nl/ min]	Over-pressure [bar]
NC 630 B	30	2,5

Adjustment of the dilution gas

- Connect the solenoid valve (C4)
- Connect the nitrogen main supply up to the connection on control panel (GC)
- Switch on the COBRA NC vacuum pump, the inlet is closed
  - ◆ Open the solenoid valve (C4) of the dilution gas circuit
- Adjust the dilution gas pressure with the help of the pressure regulator (C1) to 2,5 bar (relative)
  - ◆ The gauge (C2) shows a visual display of the dilution gas pressure
- Adjust the dilution gas volume flow with the help of the regulating valve of the flow meter (C3)
  - ◆ The dilution gas volume flow must correspond to the set flow value
  - ◆ The dilution gas volume flow depends also on the application
- Close the solenoid valve (C4) of the dilution gas circuit

Barrier gas, without control panel

Connection



GC	Gas connection G 1/4
D1	Pressure regulator
D2	Manometer
D3	Flow meter
D4	Needle valve

Set values

Type	Volume flow [Nl/ min]	Over-pressure [bar]
NC 630 B	5 ... 5,75	1

Adjustment of the barrier gas:

- Connect the nitrogen main supply up to the connection on the control panel (GC)
  - ◆ Over-pressure: 1 bar
  - ◆ Gas type: nitrogen or compressed air
- Adjust the barrier gas pressure with the help of the pressure regulator (D1) to 1 bar (relative)
  - ◆ The gauge (D2) shows a visual display of the barrier gas pressure
- Adjust barrier gas volume flow with the help of the needle valve (D4) between 5 and 5,75 Nl/ min
  - ◆ The flow meter (D3) shows a visual display of the barrier gas volume flow

NOTE: The barrier gas volume flow should not drop below the set flow value.

Saving the operating parameters

As soon as the vacuum pump has been working for 30 minutes under normal conditions after having been switched on:

- Measure the working current of the motor and keep it as reference value for all future maintenance and repair work

Recommendations on operation

Application

WARNING

The vacuum pump is designed for operation under the conditions described below.

In case of disrespect of the above, risk of damage or destruction of the vacuum pump!

The vacuum pump must only be operated under the conditions described below.

The COBRA NC vacuum pumps have been designed for use in industrial applications and similar industries.

They can be used to draw gases and gas mixtures.

WARNING

When using toxic, inflammable and / or explosive gases, make sure that the system corresponds to applicable local and national safety regulations and that all applicable safety measures are followed. All product-specific safety regulations must be observed.

Solid particles must not enter the vacuum pump. In case of process errors, the pump can draw in a certain amount of liquids. If the pump has drawn in liquid, a short drying run will be necessary at the end of the process and, in case a silencer (v) (accessory) is mounted at the outlet:

- ◆ Drain the silencer (v) (accessory)

The vacuum pump is intended for use in a potentially non-explosive environment.

The vacuum pump is thermally suitable for continuous use at any pressure between atmospheric pressure and ultimate pressure.

CAUTION

During operation the surface of the vacuum pump can exceed temperatures of 90°C.

Risk of burns!

The vacuum pump must be protected against contact during operation. If touching the pump is unavoidable, wait until the surface temperature has cooled down or wear protective gloves.

CAUTION

The sound level of the pump within a certain perimeter of the pump is high.

Risk of hearing damage.

Users, who are spending a longer period of time in the vicinity of a non-insulated vacuum pump, must wear suitable ear protection.

CAUTION

The COBRA NC vacuum pumps are always delivered without oil.

Operation without oil will destroy the vacuum pump within a short period of time.

The vacuum pump must remain in a horizontal position once it has been filled with oil.

CAUTION

The COBRA NC vacuum pumps are always delivered without cooling liquid.

Operation without cooling liquid will destroy the vacuum pump within a short period of time!

- Make sure that all provided covers, guards, hoods etc. remain fitted
- Make sure that protective devices will not be disconnected
- Make sure that there is no leakage in the system, the escape of dangerous substances must be avoided
- Make sure that the “Installation Prerequisites” are complied with and will remain so, and ensure that adequate cooling is guaranteed

If the pump is shut down for a longer period of time:

CAUTION

If there is a risk of frost, all the cooling water must be drained out of the vacuum pump if the pump is shut down for a longer period of time!

– Drain the cooling liquid
  - ◆ Open the cooling liquid vent plug (q)
  - ◆ Open the cooling liquid purge ball valve (r)
  - ◆ Drain the cooling liquid
  - ◆ Refit the cooling liquid vent plug again
  - ◆ Refit the cooling liquid vent plug again
  - ◆ Close the cooling liquid purge ball valve
  - ◆ Collect the cooling liquid and re-use it or dispose of it according to local or national regulations

– Drain the cooling water
  - ◆ Disconnect the cooling water inlet / outlet connections
  - ◆ Drain the cooling water completely
  - ◆ Drain the cooling water with the help of compressed air in order to prevent any risk of frost or corrosion

NOTE: When the pump has not been in operation for a few days or when a sticky substance has been drawn, it is possible that the two rotor screws of the COBRA NC vacuum pump stick to each other. Unscrew the access plug (k) to the rotor screws. Loosen the rotors from each other with an Allen key and an extension by turning them clockwise by hand.

Switching the vacuum pump on / off

First start-up of the system

- Make sure that the “Installation Prerequisites” are followed
- Check the position of the thermostatic valve

When the thermostatic valve is in position 0 (cold position):

- ◆ The thermostatic valve is open

When the thermostatic valve is in position 3-3,5 (hot position):

- ◆ The thermostatic valve is closed

If the system is equipped with a solenoid valve on the cooling water circuit:

- ◆ Open the solenoid valve

- Open the cooling water supply

If the vacuum pump is equipped with a barrier gas system:

- ◆ Open the solenoid valve
- ◆ Open the nitrogen supply
- ◆ Adjust the barrier gas pressure

If the vacuum pump is equipped with a purge gas system:

- Open the solenoid valve
- Open the nitrogen supply
- Adjust the pressure and volume flow for the purge gas

If the vacuum pump is equipped with a dilution gas system:

- Open the solenoid valve
- Open the nitrogen supply
- Adjust the pressure and volume flow for the dilution gas

- Start the vacuum pump

If the vacuum pump is equipped with a solenoid valve at the inlet:

- Open the solenoid valve

If the vacuum pump is equipped with a shut-off valve at the inlet:

- Open the shut-off valve

Switching off the system

If the vacuum pump is equipped with a solenoid valve at the inlet:

- Close the solenoid valve

If the vacuum pump is equipped with a shut-off valve at the inlet:

- Close the shut-off valve

If the vacuum pump is equipped with a purge gas system:

- Open the solenoid valve on the flushing device
- Flush the vacuum pump for 20 - 40 minutes
- Close the solenoid valve on the flushing device

- Switch off the vacuum pump
- Close the cooling water supply

If the vacuum pump is equipped with a solenoid valve on the cooling water circuit:

- Close the solenoid valve

If the vacuum pump is equipped with a barrier gas, purge or dilution gas system:

- Close the nitrogen supply
- Close the solenoid valve(s)

- The system must be disconnected from the power supply

Maintenance

DANGER

In case the vacuum pump has conveyed gases that have been contaminated with foreign materials that are dangerous to health, the oil and condensates will also be contaminated.

These foreign materials can infiltrate the pores, recesses and other internal spaces of the vacuum pump.

Danger to health when the vacuum pump is dismantled.

Danger to the environment.

Always wear protective clothing when carrying out maintenance work.

Before any maintenance work, the inlet and outlet piping as well as the vacuum pump itself must be flushed with nitrogen.

CAUTION

Only authorised personnel may carry out any dismantling on the vacuum pump. Before work begins, the operator of the vacuum pump must fill in a form or a "Declaration Regarding Contamination of Vacuum Equipment and Components" that provides information on possible dangers and appropriate measures.
If this form has not been filled in completely and signed by a responsible person, the vacuum pump may not be dismantled.

CAUTION

Before any maintenance work is started, a safety perimeter of at least 1 [m] around the machine must be set up.

CAUTION

During operation the surface of the vacuum pump may reach temperatures in excess of 90 °C.

Risk of burns!

Before starting any maintenance work, make sure that the vacuum pump has been fully switched off and that it cannot accidentally be switched on again. Follow the shutdown procedure in the section "Stopping procedure for maintenance":

- Stop the vacuum pump
- Switch off the power supply (the vacuum pump must be fully disconnected from the power supply)
- Disconnect the cooling water connections (inlet first, then outlet)
- Put up label or warning board "Maintenance in progress" on or next to the pump

CAUTION

The oil temperature can reach a value of 100°C!

Danger of burns!

- Make sure that the oil circuit and the cooling liquid circuit have been drained before moving the vacuum pump
- Make sure that any cleaning materials used to clean the vacuum pumps have been disposed of according to local and national regulations

Before disconnecting the different connections, make sure that the inlet and exhaust lines of the vacuum pump have been brought to atmospheric pressure

When the maintenance work has been finished, follow the procedure "Start-up procedure after maintenance":

- Remove the label or warning board "Maintenance in progress"
- Check the cooling liquid and oil levels according to chapters "Checking the oil level" and "Checking the cooling liquid level"
- Connect the pump up to the power supply
- Reconnect the cooling water connections (outlet first, then inlet)
- Make sure that the "Installation Prerequisites" are followed
- Start the pump

Maintenance program

NOTE: The maintenance intervals depend on the individual operating conditions. The intervals given below should be considered as initial guidelines which should be shortened or extended as appropriate. In particularly heavy duty operation such as high dust loads in the environment or in the process gas, it can become necessary to shorten the maintenance intervals significantly.

Monthly:

- Check the oil level (see "Checking the Oil")
- Check the level of the cooling liquid (see "Checking the Cooling liquid")
- Check the temperature of the cooling liquid (see "Checking the Cooling water")
- Check the vacuum pump for oil leaks - in case of leaks, have the vacuum pump repaired (Busch service)
- Check the vacuum pump for cooling liquid leaks - in case of leaks, have the vacuum pump repaired (Busch service)
- Check the vacuum pump for cooling water leaks - in case of leaks, have the vacuum pump repaired (Busch service)

Yearly:

In case of operation in a dusty environment:

- ◆ Make sure that the operating room is clean and free of dust; clean the room if necessary

- Make sure that the vacuum pump has been switched off and that it cannot accidentally be switched on again
- Check the electrical connections
- Carry out a visual inspection of the vacuum pump
- Check the operation of the gas ballast assembly (s) (option), disassemble and clean as required. In the case of excessive discolouration of the inlet filter change this item

If the inlet is equipped with a mesh screen:

- ◆ Check the mesh screen at the inlet and clean it if necessary

- Check the correct operation of the measurement and safety equipment

Every 1000 operating hours:

If the discharge is equipped with a silencer:

- ◆ Bleed the condensation of the silencer through the purge system

Every 5000 operating hours:

- Change the oil (see "Changing the Oil")
- Change the cooling liquid (see "Changing the cooling liquid")

If an filter is installed on the cooling water line:

- ◆ Check the filter, clean or replace it, if necessary

Every 10'000 operating hours:

- Check the seals and replace them if necessary
- Check the inlet and discharge lines and clean or replace them if necessary

If the vacuum pump is equipped with oil-lubricated single mechanical seals:

- ◆ Check the external look of the mechanical seals. If they are worn or damaged, replace them.

Every 16'000 operating hours, at the latest after 4 Years:

Have a major overhaul done on the vacuum pump (Busch service)

Stopping procedure for maintenance

- Stop the vacuum pump
- Switch off the power supply (the vacuum pump must be disconnected from the power supply)
- Disconnect the cooling water connections (inlet first, then outlet)
- Put up label or warning board "Maintenance in progress" on or next to the pump.

Start-up after maintenance

- Remove label or warning board "Maintenance in progress"
- Check the cooling liquid and oil levels according to the chapters "Checking the oil level" and "Checking the cooling liquid level"
- Connect the pump up to the power supply
- Reconnect the cooling water connections (outlet first, then inlet)
- Make sure that the "Necessary installation instructions" are followed
- Start the vacuum pump

Checking the oil

CAUTION

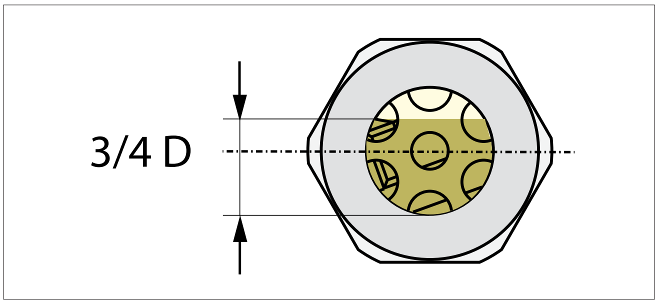
It is possible that the vacuum pump was tested with an oil type different that that will be used for your application. The vacuum pumps tested with a special oil are dispatched with specific stickers ("Special Oil") on the gear box housing like on the cylinder cover inlet side (B-side).

If the oil type is not compatible, want to carry out a cleaning of the parts in contact with oil.

Make sure that the bearings are lubricated before the reassembly.

Checking the oil level

- Make sure that the vacuum pump is switched off and cannot accidentally be switched on again
- Check the oil level on the different oil sight glasses
- The required oil level is in the top quarter of the oil sight glasses (d)



- If the oil level is below the required level:
  - ◆ Top up with oil (see "Topping up Oil")

If the oil level is above the required level:

- ◆ Check the evacuation of the condensates.
- ◆ Drain the oil (see "Draining oil")

Topping up with oil

NOTE: Under normal conditions there should be no need to top up with oil in-between the recommended oil change intervals. A significant drop in the oil level indicates a malfunction (see "Troubleshooting").

CAUTION

Fill in oil only through the oil filler holes.

CAUTION

Risk of injury (burns) with open oil filler orifice.

Risk of injury in case of badly screwed-in plugs.

Remove the oil filler plugs only when the vacuum pump is stopped.

The vacuum pump must only be operated when the oil filler plugs are firmly tightened up and do not leak.

- Make sure that the vacuum pump is switched off and cannot accidentally be switched on again
- Remove the oil filler plugs (c)
- Top up with oil
- Make sure that the oil level still is in the top quarter of the oil sight glasses (d)
- Make sure that the seals of the oil filler plugs are not damaged and replace plugs if necessary
- Refit the oil filler plugs and tighten up

Life-span of the oil

The oil life depends very much on the operating conditions. Under normal conditions the oil must be changed every 5000 operating hours or after six months, whichever comes first.

Under very unfavourable operating conditions the oil life can be less than 500 operating hours. A very short oil life time indicates either a malfunction (see "troubleshooting"9 or unsuitable operating conditions.

If there is no experience available regarding the oil life under the prevailing operation conditions, it is recomended to have the oil analysed every 500 operating hours and thus to establish the oil change interval accordingly.

Changing the oil

WARNING

In case the vacuum pump has conveyed gases that have been contaminated with harmful foreign materials, the oil will also be contaminated.

Danger to health during the change of contaminated oil.

Danger to the environment.

Wear protective equipment during the change of contaminated oil.

Contaminated oil is hazardous waste and must be disposed of separately in compliance with applicable regulations.

Draining oil

NOTE: After switching off the vacuum pump at normal operating temperature wait no more than 20 minutes before the oil is drained.

- Make sure that the vacuum pump is switched off and cannot accidentally be switched on again
- Make sure that the vacuum pump is vented to atmospheric pressure
- Put a drain tray underneath the oil drain plugs (e)
- Remove the oil drain plugs (e)
- Carefully remove the drain plugs
- Drain the oil
- Because of wear and tear on the seals replace the current drain plugs with new ones

When the oil flow has stopped:

- Close the oil drain plugs (e)
- Switch on the vacuum pump for a few seconds
- Make sure that the vacuum pump is switched off and cannot accidentally be switched on again
- Remove the oil drain plugs again and drain any remaining oil
- Check that no metal swarf sticks to the magnet of the drain plug, clean if necessary
- Refit the oil drain plugs and tighten up
- Dispose of the used oil in compliance with applicable regulations

CAUTION

Because the ends of the drain plugs are magnetic, metal swarf can stick to them. Always clean away this swarf when removing the drain plugs.

Because of wear and tear of the seals, it is recommended to replace the drain plugs whenever the oil is changed.

Filling in new oil

- Prepare the quantity of oil needed (see “Oil type/quantity”)

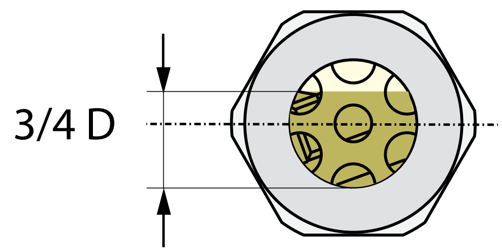
NOTE: The quantity of oil specified in the operating instructions is of informative nature only. Check the oil level with the help of the various oil sight glasses on the vacuum pump.

- Make sure that the oil drain plugs have been fitted properly and that they don’t leak
- Make sure that the oil drain plugs have been fitted properly and that they don’t leak

CAUTION

Only fill in oil through the oil filler orifices.

- Remove the oil filler plugs (c)
- Fill in oil
- Make sure that the oil level still is in the top quarter of the oil sight glasses (d)



- Make sure that the seals of the oil filler plugs are not damaged and replace plugs if necessary
- Refit the oil filler plugs and tighten up

Checking the cooling liquid

Checking the level of the cooling liquid

- Make sure that the vacuum pump is switched off and cannot accidentally be switched on again
- Check the level of the cooling liquid
- Open the cooling liquid vent plug (q) on cylinder upper plate
  - ◆ The filling level must be just under the cylinder upper plate

If the level is below:

  - ◆ Top up with cooling liquid (see “Refilling cooling liquid”)

Topping up with cooling liquid

NOTE: Under normal conditions there should be no need to top up with cooling liquid in-between the recommended change intervals. A significant drop in the cooling liquid level indicates a malfunction (see "Troubleshooting").

- Make sure that the vacuum pump is switched off and cannot accidentally be switched on again

> CAUTION
>
> Wait until pump has cooled down to ambient temperature

- Remove the cooling liquid filler plug (p) on the cylinder
- Open the cooling liquid drain plug (q1) on cylinder upper plate
- Fill in cooling liquid by filler hole until the cooling liquid flows of the cooling vent plug (q) on cylinder upper plate
- Close the cooling liquid vent plug (q) on cylinder upper plate
- Fit the cooling liquid filler plug (p) on cylinder upper plate
- Make sure that the seal rings of the cooling liquid filler plug and cooling liquid drain plugs are not damaged, replace plugs if necessary
- If there is any spillage on the surface of the vacuum pump, wipe it up

Changing the cooling liquid

Draining the cooling liquid

- Make sure that the vacuum pump is switched off and cannot accidentally be switched on again

> CAUTION
>
> Wait until pump has cooled down to ambient temperature

- Make sure that the vacuum pump is vented to atmospheric pressure
- Put a drain tray underneath the cooling liquid purge ball valve (r)
- Remove the cooling liquid vent plug (q)
- Open the cooling liquid purge ball valve (r)
- Drain the cooling liquid
- Refit the cooling liquid drain plug (q1) and tighten up
- Close the cooling liquid purge ball valve (r)
- Collect the cooling liquid and re-use it or dispose of it according to local or national regulations

Filling up with new cooling liquid

- Prepare the quantity of cooling liquid needed (see "Cooling liquid type / quantity")

NOTE: The quantity of cooling liquid specified in the operating instructions is of informative nature only.

- Remove the cooling liquid filler plug (p) on cylinder upper plate
- Open the cooling liquid drain plug (q1) on cylinder upper plate
- Close the cooling liquid vent plug (q) on cylinder upper plate
- Fit the cooling liquid filler plug (p) on cylinder upper plate
- If there is any spillage on the surface of the vacuum pump, wipe it up
- Start the vacuum pump

If the inlet line is equipped with a shut-off device:

- ◆ Close the shut-off device

If the inlet line is not equipped with a shut-off device:

- ◆ Place a rubber mat on the suction flange

- Let the vacuum pump run for maximum 5 minutes
- Stop the vacuum pump and wait for a few minutes
- Open the cooling liquid vent plug (q) on cylinder upper plate
- Check that the filling level is just under the cylinder upper plate

In case the cooling liquid level is below thhe required level:

- ◆ Fill in more cooling liquid

If the inlet line is equipped with a shut-off device:

- ◆ Open the shut-off device

If the inlet line is not equipped with a shut-off device:

- ◆ Remove the rubber mat from the inlet flange and connect the inlet line to the inlet flange

Checking the cooling water

Checking the current consumption

- Check the current of the motor

An increased current indicates a fault (see "Troubleshooting")

Checking the silencer (v) (accessory)

- Make sure that the condensates do not collect at the outlet of the vacuum pump
- Drain the condensates via the drain provided and collect them in a container
- Dispose of the condensates in compliance with applicable environmental protection regulations
- Check regularly the silencer and clean it if necessary

> CAUTION
>
> Wear protective clothing when carrying out maintenance work on the silencer.
>
> Risk of contaminated process residues.

Checking the gas ballast (s) (option)

Check the operation of the gas ballast assembly, disassemble and clean as required. In the case of excessive discolouration of the inlet filter change this item

Overhaul

> CAUTION
>
> Improper maintenance work on the vacuum pump can damage it.
>
> Risk of explosion!
>
> Non-adherence to the procedure will cancel approval for start-up of the pump !
>
> Any dismantling of the vacuum pump beyond of what is described in this manual must be done by specially trained Busch service staff only.

DANGER

In case the vacuum pump has conveyed gases that have been contaminated with foreign materials that are dangerous to health, the oil and condensates will also be contaminated.

These foreign materials can infiltrate the pores, recesses and other internal spaces of the vacuum pump.

Danger to health when the vacuum pump is dismantled.

Danger to the environment.

Prior to shipping, the vacuum pump must imperatively be decontaminated and the degree of contamination must be documented in a declaration of decontamination ("Declaration of Decontamination"), which can be downloaded from www.busch-vacuum.com.

Busch service will only accept vacuum pumps that come with a completely filled in and legally binding signed form.

Removal from service

Temporary removal from service

Prior to disconnecting inlet and outlet pipes as well as cooling water pipes, make sure that all piping is vented to atmospheric pressure

Recommissioning

CAUTION

After a long period of inactivity, it is possible that the rotor screws of the COBRA NC vacuum pump are stuck.

Turn the rotor screws manually

- Make sure that all gaskets, plugs or adhesive tapes have been removed.
- Start the vacuum pump as described in the chapter "Installation and Commissioning"

Dismantling and Disposal of the vacuum pump

DANGER

In case the vacuum pump has conveyed gases that have been contaminated with harmful foreign material which are harmful to health, the oil and the condensates will also be contaminated with harmful foreign material.

These foreign materials can infiltrate the pores, recesses and other internal spaces of the vacuum pump.

Danger to health during dismantling of the vacuum pump.

Danger to the environment.

During dismantling of the vacuum pump protective equipment and clothing must be worn.

The vacuum pump must be decontaminated prior to disposal.

Prior to shipping, the vacuum pump must imperatively be decontaminated and the degree of contamination must be documented in a declaration of decontamination ("Declaration of Decontamination"), which can be downloaded from www.busch-vacuum.com.

Used oil and condensates must be disposed of separately in compliance with applicable environmental regulations.

When the vacuum pump comes to the end of its life:

– it must be decontaminated

CAUTION

Only authorised personnel may carry out dismantling work on the vacuum pump. Before work begins, the operator of the vacuum pump must fill in a form or a "Declaration of Decontamination" that provides information on possible dangers and appropriate measures.

If this form has not been filled in completely and signed, the vacuum pump may not be dismantled.

– drain the oil
  - ◆ dispose of the used oil in compliance with applicable environmental regulations
– drain the cooling liquid
  - ◆ dispose of the cooling liquid in compliance with applicable environmental regulations
– dismantle the vacuum pump

CAUTION

During dismantling of the vacuum pump protective equipment and clothing must be worn

  - ◆ dispose of the vacuum pump as scrap metal
– dispose of the different components of the pump in compliance with applicable regulations

Troubleshooting

WARNING

Risk of electrical shock, risk of damage to equipment.

Electrical installation work must only be executed by qualified personnel that knows and observes the following regulations:
- IEC 364 or CENELEC HD 384 or DIN VDE 0100, respectively,
- IEC-Report 664 or DIN VDE 0110,
- BGV A2 (VBG 4) or equivalent national accident prevention regulation.

CAUTION

During operation the surface of the vacuum pump may reach temperatures of more than 90 °C.

Risk of burns!

Let the vacuum pump cool down prior to a required contact or wear heat protection gloves.

Problem	Possible Cause	Remedy
The vacuum pump does not reach the usual pressure	The vacuum system or suction line is not leak-tight	Check the hose or pipe connections for possible leak
The drive motor draws a too high current (compare with initial value after commissioning))	Process deposits on the pumping components (the most common cause)	Flush the pump carefully
Evacuation of the system takes too long	In case a mesh screen is installed on the suction connection: The mesh screen on the suction connection is partly clogged	Clean the mesh screen If cleaning is required too frequently install a filter upstream
	In case an inlet filter is installed on the suction connection: The filter on the suction connection is partly clogged	Clean or replace the inlet filter, respectively
	Partial clogging in the suction, discharge or pressure line	Remove the clogging
	Long suction, discharge or pressure line with too small diameter	Use larger diameter
	Internal parts worn or damaged	Repair the vacuum pump (Busch service)
The vacuum pump does not start	The drive motor is not supplied with the correct voltage or is overloaded	Supply the drive motor with the correct voltage
	The drive motor starter overload protection is too small or trip level is too low	Compare the trip level of the drive motor starter overload protection with the data on the nameplate Correct if necessary In case of high ambient temperature: Set the trip level of the drive motor starter overload protection 5 percent above the nominal drive motor current
	One of the fuses has blown	Check the fuses
	The connection cable is too small or too long causing a voltage drop at the vacuum pump	Use sufficiently dimensioned cable
	The vacuum pump or the drive motor is blocked	Turn the screws rotors manually (k) or repair the vacuum pump (Busch service)
	The drive motor is defective	Replace the drive motor (Busch service)

The vacuum pump is blocked	Solid foreign matter has entered the vacuum pump	Repair the vacuum pump (Busch service) Make sure the suction line is equipped with a mesh screen If necessary additionally provide a filter
	Corrosion in the vacuum pump from remaining condensate	Repair the vacuum pump (Busch service) Check the process Observe the chapter “Installation and Commissioning, Operating Notes”
	The vacuum pump was run in the wrong direction	Repair the vacuum pump (Busch service) When connecting the vacuum pump make sure the vacuum pump will run in the correct direction (see “Installation”)
	Condensate ran into the vacuum pump	Repair the vacuum pump (Busch service) Make sure no condensate will enter the vacuum pump, if necessary provide a drip leg and a drain cock Drain condensate regularly
	The recirculating pump is blocked	Remove the recirculating pump, clean if necessary
The vacuum pump starts, but labours or runs noisily or rattles The drive motor draws a too high current (compare with initial value after commissioning)	Connections in the drive motor terminal box are defective Not all drive motor coils are properly connected The drive motor operates on two phases only	Check the proper connection of the wires against the connection diagram Tighten or replace loose connections
	The vacuum pump runs in the wrong direction	Verification and rectification see “Installation and Commissioning”, correct if necessary
	Standstill over several weeks or months	Let the vacuum pump run warm with inlet closed
	Oil viscosity is too high for the ambient temperature	Use oil with a better viscosity class
	Improper oil quantity, unsuitable oil type	Use the proper quantity of one of the recommended oils (see “Oil”, oil change see “Maintenance”)
	No oil change over extended period of time	Perform oil change incl. flushing (see “Maintenance”)
	Foreign objects in the vacuum pump Stuck bearings	Repair the vacuum pump (Busch service)
The vacuum pump runs very noisily	Defective bearings	Repair the vacuum pump (Busch service)
	Worn coupling element	Replace the coupling element
The vacuum pump runs very hot (the oil sump temperature shall not exceed 100 °C)	Cooling water flow too low	Check the cooling water circuit and adjust the flow if necessary
	Ambient temperature too high	Observe the permitted ambient temperatures
	Temperature of the inlet gas too high	Observe the permitted temperatures for the inlet gas
	Oil level too low	Top up oil Check the oil filler plugs Check the mechanical seals (option)
	Oil burning by an overheating of the vacuum pump	Drain oil Fill in new oil (see “Maintenance”) In case the oil life is too short: use oil with better heat resistance or retrofit cooling

	Mains frequency or voltage outside tolerance range	Provide a more stable power supply
	In case a mesh screen is installed on the suction connection: The mesh screen on the suction connection is partially clogged	Clean the mesh screen If cleaning is required too frequently install a filter upstream
	In case an inlet air filter is installed on the suction connection: The filter on the suction connection is partially clogged	Clean or replace the filter
	Partial clogging in the suction or discharge line	Remove the clogging
	Long suction, discharge or pressure line withtoo small diameter	Use larger diameter
The oil is black	Oil change intervals are too long The oil was overheated	Drain oil Fill in new oil (see “Maintenance”) In case the oil life is too short: use oil with better heat resistance or retrofit cooling

Spare parts

NOTE: When ordering spare parts or accessories acc. to the table below please always quote the type and the serial no. of the vacuum pump. This will allow Busch service to check if the vacuum pump is compatible with a modified or improved part.

This parts list applies to a typical configuration of the standard vacuum pump. Depending on the specific order deviating parts data may apply.

Pos.	Description	Qt	Article no.
9	O-ring	1	0486 000 524
380.9	O-ring	1	0486 000 640
42	Piston ring	14	0488 527 221
43	O-ring	2	0486 000 529
44	O-ring	2	0486 000 531
57	Piston ring	14	0488 527 221
58	O-ring	2	0486 000 529
59	O-ring	2	0486 000 531
63	Adjusting ring	2	0433 524 082
100	Grooved ball bearing single row	2	0473 539 779
101	Angular ball single row	2	0473 540 307
101*	Angular ball single row	2	0473 552 527
102	Angular ball single row	2	0473 539 778
102*	Angular ball single row	2	0473 550 493
103	Angular ball bearing	2	0473 539 780
130	O-ring	1	0486 539 797
131	O-ring	1	0486 539 796
132	O-ring	1	0486 539 797
133	O-ring	1	0486 541 432
134	O-ring	1	0486 539 798
135	O-ring	1	0486 539 798
136	O-ring	1	0486 537 258
137	O-ring	1	0486 539 799
138	O-ring	1	0486 000 698
139	O-ring	1	0486 000 612
140	Shaft seal	1	0487 547 735
141	Sleeve	1	0472 545 921
141	Distance ring	1	0460 000 215
146	O-ring	1	0486 000 633
147	O-ring	1	0486 000 633
188	Magnetic plug	1	0415 134 870
195	Magnetic plug	1	0415 134 870
403	Coupler sleeve	1	0512 533 649

Spare parts kit

Wearing parts	Description	Article No.
Set of seals	consisting of all necessary seals	0990 542 822
Set of seals*	consisting of all necessary seals	0990 557 013
Overhaul kit	consisting of set of seals and all wearing parts	0993 548 774
Overhaul kit*	consisting of set of seals and all wearing parts	0993 557 014

* only for PFPE "Fomblin®" oil version

Oil type/ quantity

> CAUTION
>
> It is possible that the vacuum pump was tested with a different type of oil to the type you will be using for your application. Vacuum pumps that have been tested with a special oil are labelled with specific stickers ("Special oil") on the gearbox housing, as well as the cylinder cover on the inlet side (B- side) upon despatch.
>
> If the oil type is not compatible, all parts that have come into contact with the oil must be cleaned.
>
> Make sure that the bearings are lubricated upon reassembly.

Oil type

VE 101 oil

- Make sure that the oil type corresponds to specifications:

– ANDEROL® 555, Process Compressor Lubricant, viscosity class ISO VG 100 (Busch VE101, part no 0831 000 099)

NOTE: Read technical sheet in appendix for more details on the product

PFPE oil (option)

- Make sure that the oil type corresponds to specifications:

– Fomblin® Y LVAC 25/6 (Busch PFPE VF 0250, part no 0831 000 054) (0,5 l ≅ 1 kg)

Pump labelled with this sticker:

NOTE: Read technical sheet in appendix for more details on the product

Oil quantity

The quantity of oil specified in the following table is of informative nature only. Check the oil level with the help of the various oil sight glasses on the vacuum pump.

Quantity [Liter]	Motor side (A)	Suction side (B)
NC 630 B	0,9	0,55

ANDEROL PRODUCT DATA SHEET

ANDEROL® Process/- Gas-Compressor Series

SYNTHETIC RECIPROCATING COMPRESSOR LUBRICANTS

GENERAL INFORMATION

This ANDEROL Process/Gas-Compressor Series comprises four ester-based products, with different viscosity levels ranging from ISO grade 100 through to 320. This scale allows you to select the right protection for each individual requirement, even for extreme low or high ambient temperatures and working conditions. The oxidation resistance and stable viscosity of this fluid eliminate varnish formation and separator blocking. Due to the excellent film strength and lubricity of this fluid you will experience less wear, higher efficiency and six to eight or even longer drain intervals than with mineral oils.

APPLICATIONS

This ANDEROL Series are highly recommended as long-term cylinder and crankcase lubricants for reciprocating Process/Gas-Compressors and vacuum pumps. Typical applications for each ANDEROL product within the series include:

- ANDEROL 555:
  for general applications with cylinder diameters up to 18"
- ANDEROL 755:
  or high ambient or operating temperatures (including Helium) and/or cylinder diameters over 18"
- ANDEROL 1200N:
  for applications (excl. H2) in combination with dry Nitrogen
- ANDEROL 1255:
  for applications where a high viscosity lubricant is required

GAS COMPATIBILITY

The following gases are compatible with this lubricant series:

Benzene	Furnace(crack)gas	Natural gas
Butadiene	Helium	Nitrogen
Carbon	Carbon	Hydrogen
Dioxide (dry)	Monoxide	Sulphide (dry)
Hydrogen	Propane	Synthetic gas
Sulphur Hexafluoride	Ethylene	Methane

(Information on other gases is available on request)

For more information please refer to the relevant Material Safety Data Sheet accompanying each product.

ADVANTAGES

The advantages with this lubricant series in process/gas-compressor applications include:

- Longer service live and greater protection of compressor components
- High compatibility with a wide variety of gases
- Very low vapour pressure
- High oxidation resistants
- Wide operating temperature
- Elimination of varnish lacquering and deposits
- Reduced energy consumption
- Lower oil consumption
- Very long drain intervals
- Reduced compressor maintenance
- Excellent air-release and non-foaming properties
- Greatly reduced fire and explosion hazard
- Compatibility with compressor seals, gaskets, hoses and air system components

COMPATIBILITY

ANDEROL Process/Gas-Compressor lubricants are not detrimental to seals, paint and plastics containing Viton, High Nitrile Buna N, Teflon Epoxy Paint, Oil-resistant Alkyd, Nylon, Delrin, Celcon and PBT.
Not recommended are Neoprene, SBR Rubber, Low Nitrile Buna N, Acrylic Paint, Lacquer, Polystyrene, PVC and ABS.

STORAGE AND HANDLING

ANDEROL Synthetic Lubricants are not highly flammable and generally present little hazard during handling if normal care is exercised.

PACKAGING

ANDEROL Process/Gas-Compressor Series lubricant are available in 200 and 60 litre drums, 20 litre, 5 litre and 1 litre cans. Upon request, the lubricants can be suppli- ed in 640 and 1000 litre (non)returnable containers as well as in bulk.

ANDEROL® Process/Gas-Compressor Series

LONG LIFE, HIGH PERFORMANCE SYNTHETIC LUBRICANTS FOR PROCESS/GAS-COMPRESSORS

TYPICAL PROPERTIES

ANDEROL	555	755	1200N	1255
Nominal Operating Range, °C	-15 tot 230	-15 tot 230	0 tot 230	0 tot 230
ISO-VG, ASTM D-2422	100	150	320	320
Viscosity @ bij 40 °C, mm^2/s ASTM D-445	94	143	316	313
Viscosity @ bij 100 °C, mm^2/s ASTM D-445	9,1	12,9	25,0	21,0
Viscosity index, ASTM D-2270	59	79	98	77
Density @ 15 °C, kg/m DIN 51.757	957	972	941	943
Pour Point, °C ASTM D-97	-42	-39	-36	-33
Flash Point, C.O.C. °C ASTM D-92	252	268	290	265
Autoignition Temperature, °C ASTM D-2155	410	410	425	428
Conradson Carbon Residue, % ASTM D-2155	0,02	0,02	0,05	0,05
Evaporation, 22 hrs. @ 99 °C, % ASTM D-972	< 1,0	< 0,3	< 1,0	< 1,0
Copper Strip Corrosion 3 hrs. @ 100 °C ASTM D-130	1a	1a	1a	1a
Four-ball Wear, 1200 rpm, 40 kg for 1 hr @ 75 °C, mm ASTM D-4172	0,8	0,8	0,4	0,37

555SER-E-050298-2

ANDEROL Synthetic Lubricants

ANDEROL BV
P.O.Box 1489, 6201 BL Maastricht
Punterweg 21A, 6222 NW Maastricht
The Netherlands
Tel: +31 (0)43 352 41 90
Fax:+31 (0)43 352 41 99

Revision Date 29.11.2010

FOMBLIN® Y LVAC

1. PRODUCT AND COMPANY IDENTIFICATION

1.1. Product identifiers

- Product name	:	**FOMBLIN® Y LVAC**
- Product grade(s)	:	06/6; 14/6; 16/6; 25/6; 300
- Chemical characterization	:	Perfluorinated polyethers
- Structural formula	:	CF3-O-(C3F6O)n-(CF2-O)m-CF3

1.2. Identified uses / Uses advised against

- Identified uses :
  - Electronic industry
  - Electrical industry
  - Chemical industry
  - For industrial use only.

1.3. Manufacturer or supplier's details

- Company	:	SOLVAY SOLEXIS S.p.A.
- Address	:	VIALE LOMBARDIA, 20 I- 20021 BOLLATE
- Telephone	:	+390238351
- Fax	:	+390238352614
- E-mail address	:	sds.solexis@solvay.com

1.4. Emergency telephone number

- Emergency telephone number **+44(0)1235 239 670 [CareChem 24] (Europe)**

2. HAZARDS IDENTIFICATION

2.1. GHS Classification

2.1.1. European regulation (EC) 1272/2008, as amended

Not classified as hazardous according to the European regulation (EC) 1272/2008, as amended

2.1.2. European Directive 67/548/EEC or 1999/45/EC, as amended

Not classified as hazardous according to European Directive 67/548/EEC or 1999/45/EC, as amended

2.2. EC Label - According to Regulation (EC) 1272/2008, as amended

No labelling

2.3. Other hazards which do not result in classification

- Thermal decomposition can lead to release of toxic and corrosive gases.

3. COMPOSITION/INFORMATION ON INGREDIENTS

3.1. Concentration

Substance name:	Concentration
1-Propene, 1,1,2,3,3,3-hexafluoro-, oxidized, polymd. CAS-No.: 69991-67-9 / EC-No.: - / Index-No.: -	> 99.9 %

4. FIRST AID MEASURES

4.1. Description of necessary first-aid measures

4.1.1. If inhaled

- Move to fresh air in case of accidental inhalation of fumes from overheating or combustion.
- Oxygen or artificial respiration if needed.

4.1.2. In case of eye contact

- Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes.
- If eye irritation persists, consult a specialist.

4.1.3. In case of skin contact

- Wash off with soap and water.
- If symptoms persist, call a physician.

4.1.4. If swallowed

- Drink 1 or 2 glasses of water.
- Do NOT induce vomiting.
- If symptoms persist, call a physician.

4.2. Most important symptoms/effects, acute and delayed

4.2.1. Inhalation

- No known effect.

4.2.2. Skin contact

- Redness

4.2.3. Eye contact

- Contact with eyes may cause irritation.
- Redness

4.2.4. Ingestion

- Ingestion may provoke the following symptoms:
- Symptoms: Nausea, Vomiting, Diarrhoea

5. FIRE-FIGHTING MEASURES

5.1. Extinguishing media

5.1.1. Suitable extinguishing media

- Water
- powder
- Foam
- Dry chemical
- Carbon dioxide (CO2)

5.1.2. Unsuitable extinguishing media

- None.

5.2. Specific hazards arising from the chemical

- The product is not flammable.
- Not explosive
- In case of fire hazardous decomposition products may be produced such as: Gaseous hydrogen fluoride (HF), Fluorophosgene

5.3. Special protective actions for fire-fighters

- Wear self-contained breathing apparatus and protective suit.
- When intervention in close proximity wear acid resistant over suit.
- Evacuate personnel to safe areas.
- Approach from upwind.
- Protect intervention team with a water spray as they approach the fire.
- Keep containers and surroundings cool with water spray.
- Keep product and empty container away from heat and sources of ignition.

6. ACCIDENTAL RELEASE MEASURES

6.1. Personal precautions, protective equipment and emergency procedures

6.1.1. Advice for non-emergency personnel

- Prevent further leakage or spillage if safe to do so.

6.1.2. Advice for emergency responders

- Ensure adequate ventilation.
- Sweep up to prevent slipping hazard.
- Keep away from open flames, hot surfaces and sources of ignition.
- Refer to protective measures listed in sections 7 and 8.

6.2. Environmental precautions

- Should not be released into the environment.
- Do not flush into surface water or sanitary sewer system.

6.3. Methods and materials for containment and cleaning up

- Soak up with inert absorbent material.
- Suitable material for picking up
- Dry sand
- Earth
- Shovel into suitable container for disposal.

7. HANDLING AND STORAGE

7.1. Precautions for safe handling

- No special handling advice required.
- Ensure adequate ventilation.
- Use personal protective equipment.
- Keep away from heat and sources of ignition.
- To avoid thermal decomposition, do not overheat.

7.2. Conditions for storage, including incompatibilities

7.2.1. Storage

- No special storage conditions required.
- Keep away from heat and sources of ignition.
- Keep in properly labelled containers.
- Keep away from combustible material.
- Keep away from incompatible products
- Provide tight electrical equipment well protected against corrosion.
- Refer to protective measures listed in sections 7 and 8.

7.2.2. Packaging material

7.2.2.1. Suitable material

- Polyethylene

7.3. Specific use(s)

- For further information, please contact: Supplier

8. EXPOSURE CONTROLS/PERSONAL PROTECTION

8.1. Control parameters

8.1.1. Exposure Limit Values

Remarks:

- Threshold limit values of by-products from thermal decomposition

Hydrogen fluoride anhydrous

- UK. EH40 Workplace Exposure Limits (WELs) 2007
  time weighted average = 1.8 ppm
  time weighted average = 1.5 mg/m3
  Remarks: as F
- UK. EH40 Workplace Exposure Limits (WELs) 2007
  Short term exposure limit = 3 ppm
  Short term exposure limit = 2.5 mg/m3

Remarks: as F
- US. ACGIH Threshold Limit Values 2009
  time weighted average = 0.5 ppm
  Remarks: as F
- US. ACGIH Threshold Limit Values 2009
  Ceiling Limit Value = 2 ppm
  Remarks: as F
- EU. Indicative Exposure and Directives relating to the protection of risks related to work exposure to chemical, physical, and biological agents. 12 2009
  time weighted average = 1.8 ppm
  time weighted average = 1.5 mg/m3
  Remarks: Indicative
- EU. Indicative Exposure and Directives relating to the protection of risks related to work exposure to chemical, physical, and biological agents. 12 2009
  Short term exposure limit = 3 ppm
  Short term exposure limit = 2.5 mg/m3
  Remarks: Indicative
- US. ACGIH Threshold Limit Values 2009
  Remarks: as F, Can be absorbed through skin.

Carbonyl difluoride
- US. ACGIH Threshold Limit Values 2009
  time weighted average = 2 ppm
- US. ACGIH Threshold Limit Values 2009
  Short term exposure limit = 5 ppm
- UK. EH40 Workplace Exposure Limits (WELs) 2007
  time weighted average = 2.5 mg/m3
  Remarks: as F
- EU. Indicative Exposure and Directives relating to the protection of risks related to work exposure to chemical, physical, and biological agents. 12 2009
  time weighted average = 2.5 mg/m3
  Remarks: Indicative

8.2. Exposure controls

8.2.1. Appropriate engineering controls
- Provide local ventilation appropriate to the product decomposition risk (see section 10).
- Refer to protective measures listed in sections 7 and 8.
- Apply technical measures to comply with the occupational exposure limits.

8.2.2. Individual protection measures

8.2.2.1. Respiratory protection
- No personal respiratory protective equipment normally required.
- Use respirator when performing operations involving potential exposure to vapour of the product.
- Use only respiratory protection that conforms to international/ national standards.

8.2.2.2. Hand protection
- Rubber or plastic gloves
- Latex gloves
- Take note of the information given by the producer concerning permeability and break through times, and of special workplace conditions (mechanical strain, duration of contact).

8.2.2.3. Eye protection
- Tightly fitting safety goggles

8.2.2.4. Skin and body protection
- Long sleeved clothing

8.2.2.5. Hygiene measures
- Ensure that eyewash stations and safety showers are close to the workstation location.
- When using do not eat, drink or smoke.
- Wash hands before breaks and at the end of workday.
- Handle in accordance with good industrial hygiene and safety practice.

8.2.3. Environmental exposure controls
- Dispose of rinse water in accordance with local and national regulations.

9. PHYSICAL AND CHEMICAL PROPERTIES

9.1. Physical and chemical properties

9.1.1. General Information

■ **Appearance**	liquid
■ **Colour**	colourless
■ **Odour**	odourless
■ **Molecular Weight**	Range of values: 1,800 - 3,300

9.1.2. Important health safety and environmental information

■ **pH**	No data
■ **pKa**	No data
■ **Melting point/freezing point**	not applicable
■ **Boiling point/boiling range**	> 270 °C
■ **Flash point**	The product is not flammable.
■ **Evaporation rate**	No data
■ **Flammability (solid, gas)**	No data
■ **Flammability**	The product is not flammable.
■ **Explosive properties**	Not explosive
■ **Vapour pressure**	0.0000001 - 0.000013 hPa, at 20 °C
■ **Vapour density**	No data
■ **Density**	1.88 - 1.90 g/cm3
■ **Relative density**	No data
■ **Bulk density**	No data
■ **Solubility(ies)**	insoluble, Water
	soluble, fluorinated solvents
■ **Solubility/qualitative**	No data
■ **Partition coefficient: n-octanol/water**	No data
■ **Autoignition temperature**	No data
■ **Decomposition temperature**	> 290 °C
■ **Viscosity**	ca. 95 - 560 mPa.s, at 20 °C
■ **Oxidizing properties**	Non oxidizer

10. STABILITY AND REACTIVITY

10.1. Reactivity

- No dangerous reaction known under conditions of normal use.

10.2. Chemical stability

- Stable under recommended storage conditions.

10.3. Possibility of hazardous reactions

- No dangerous reaction known under conditions of normal use.

10.4. Conditions to avoid

- To avoid thermal decomposition, do not overheat.
- Keep away from flames and sparks.

10.5. Materials to avoid

- Lewis acids (Friedel-Crafts) above 100°C
- Aluminum and magnesium in powder form above 200°C
- metals promote and lower decomposition temperature

10.6. Hazardous decomposition products

- Gaseous hydrogen fluoride (HF)., Fluorophosgene

11. TOXICOLOGICAL INFORMATION

11.1. Acute toxicity

11.1.1. Acute oral toxicity

- LD50, rat, > 15,000 mg/kg

11.1.2. Acute inhalation toxicity

- no data available

11.1.3. Acute dermal toxicity

- LD50, rat, > 5,000 mg/kg

11.2. Skin corrosion/irritation

- rabbit, No skin irritation
- rabbit, No skin irritation, 14 days

11.3. Serious eye damage/eye irritation

- rabbit, No eye irritation

11.4. Respiratory or skin sensitization

- guinea pig, Did not cause sensitization on laboratory animals., Skin

11.5. Mutagenicity

- Not mutagenic in Ames Test.
- negative, Chromosome aberration test in vitro

11.6. Carcinogenicity

- no data available

11.7. Toxicity for reproduction

- no data available

11.8. Specific target organ toxicity - single exposure

- Remarks: no data available

11.9. Specific target organ toxicity - repeated exposure

- Remarks: no data available

11.10. Aspiration hazard

- no data available

11.11. Other information

- Description of possible hazardous to health effects is based on experience and/or toxicological characteristics of several components.
- The product is biologically inert.
- Thermal decomposition can lead to release of toxic and corrosive gases.
- Exposure to decomposition products
- Causes severe irritation of eyes, skin and mucous membranes.

12. ECOLOGICAL INFORMATION

12.1. Toxicity

- Fishes, Brachydanio rerio, LC50, 96 h, > 360 mg/l, saturated aqueous solution
- Daphnia magna (Water flea), EC50, 48 h, > 360 mg/l, saturated aqueous solution

12.2. Persistence and degradability

12.2.1. Abiotic degradation

- Result: no data available

12.2.2. Biodegradation

- no data available

12.3. Bioaccumulative potential

- Result: no data available

12.4. Mobility

- no data available

12.5. Other adverse effects

- Ecological injuries are not known or expected under normal use.

13. DISPOSAL CONSIDERATIONS

13.1. Waste disposal methods

- Can be incinerated, when in compliance with local regulations.
- The incinerator must be equipped with a system for the neutralisation or recovery of HF.
- In accordance with local and national regulations.

13.2. Contaminated packaging

- Empty containers can be landfilled, when in accordance with the local regulations.

14. TRANSPORT INFORMATION

14.1. International transport regulations

- Sea (IMO/IMDG)
- not regulated
- Air (ICAO/IATA)
- not regulated
- European Road/Rail (ADR/RID)
- not regulated
- Inland waterway transport
- not regulated

15. REGULATORY INFORMATION

15.1. Applicable Laws or Regulations

- Regulation (EC) No 1272/2008 of the European Parliament and of the Council of 16 December 2008 on classification, labelling and packaging of substances and mixtures, as amended
- Regulation (EC) No 1907/2006 of the European Parliament and of the Council of 18 December 2006 concerning the Registration, Evaluation, Authorisation and Restriction of Chemicals (REACH), as amended
- European Waste Catalogue
- Waste codes should be assigned by the user based on the application for which the product was used.

15.2. Notification status

Inventory Information	Status
Toxic Substance Control Act list (TSCA)	- In compliance with inventory
Australian Inventory of Chemical Substances (AICS)	- In compliance with inventory
Canadian Domestic Substances List (DSL)	- In compliance with inventory
Inventory of Existing Chemical Substances (China) (IECS)	- In compliance with inventory
Korean Existing Chemicals Inventory (KECI (KR))	- In compliance with inventory
Japanese Existing and New Chemical Substances (MITI List) (ENCS)	- In compliance with inventory
New Zealand Inventory (in preparation) (NZ)	- In compliance with inventory
Philippine Inventory of Chemicals and Chemical Substances (PICCS)	- In compliance with inventory
EU list of existing chemical substances (EINECS)	- not applicable - Product falls under the EU-polymer definition.

16. OTHER INFORMATION

16.1. Other information

- New (MSDS)
- Distribute new edition to clients

This SDS is only intended for the indicated country to which it is applicable. The European SDS format compliant with the applicable European legislation is not intended for use nor distribution in countries outside the European Union with the exception of Norway and Switzerland. Safety datasheets applicable in other countries/regions are available upon request. The information given corresponds to the current state of our knowledge and experience of the product, and is not exhaustive. This applies to product which conforms to the specification, unless otherwise stated. In this case of combinations and mixtures one must make sure that no new dangers can arise. In any case, the user is not exempt from observing all legal, administrative and regulatory procedures relating to the product, personal hygiene, and protection of human welfare and the environment.

Print Date: 02.12.2010

Cooling liquid type/ quantity

Cooling liquid type

- Make sure that the cooling liquid type corresponds to specifications :

– Glycol (Ethylene), part no 0831 558 702, Zitrec MC

NOTE: Read technical sheet in appendix for more details on the product

Cooling liquid quantity

The quantity of cooling liquid specified in this instructions manual is of informative nature only. Respect the procedure of filling the cooling liquid.

29 l - proportion 40% Glycol, 60% water

Zitrec
Heat transfer fluids

PRODUCT INFORMATION

Zitrec™ MC

DESCRIPTION

Zitrec MC - mixed with the appropriate amount of water - is used as a multipurpose heat transfer fluid based on mono ethylene glycol.

APPLICATION

Many applications in the industry require a fluid to transport heat or cold. Those applications range from solar panels or heat pump systems, over cooling or heating of industrial processes and refrigerants in indirect cooling systems to artificial ski-tracks or ice rigs. This transport medium is usually called secondary refrigerant or secondary coolant. The ideal secondary refrigerant must ensure a good thermal conductivity, have a high specific heat and low viscosity. It is also important that the secondary refrigerant is non-flammable and compatible with common engineering materials.

Zitrec MC provides protection against boiling, freezing and corrosion. The dilution is determined by system requirements, mainly freezing requirements. However, to ensure good corrosion protection it is recommended to use at least 33 vol. % of **Zitrec MC** in the coolant solution, which provides freeze protection to -20°C. For lower freezing protection it is recommended to use **Zitrec M –15°C**. This ready-to-use solution contains an adjusted corrosion inhibitor package to ensure optimal corrosion protection.
Mixtures with more than 70 vol. % of **Zitrec MC** in water are not recommended, because the freeze point is increasing again and physical properties are worse.

Dilution **Zitrec M**, vol %	Freeze Point, °C	Dilution **Zitrec M**, vol %	Freeze Point, °C
28.0	- 15	52.4	- 40
39.1	- 25	56.2	- 45
43.8	- 30	63.5	- 55

COMPATIBILITY AND MIXABILITY

Zitrec MC is compatible with most other heat transfer fluids based on ethylene glycol. Exclusive use of **Zitrec MC** is recommended for optimal corrosion protection. This heat transfer fluid is compatible with European hard tap waters, up to a water hardness of 30 °dH (German hardness degrees, equivalent to 535 mg/l $CaCO_3$).

ARTECO
A Joint Venture Company

Zitrec MC

CHEMICAL AND PHYSICAL PROPERTIES

Properties	Zitrec MC	Method	Properties	Zitrec MC	Method
Ethylene glycol	92 % w/w glycol	internal			
Inhibitor content	5 % w/w	internal	Specific gravity, 20°C	1.113 typ.	ASTM D1122
Water content	5 % w/w max	ASTM D1123	Equilibrium boiling point	180°C typ.	ASTM D1120
Nitrite, amine, phosphate	nil	IC	pH	8.6 typ.	ASTM D1287
Colour	yellow	visual	Refractive Index, 20°C	1.431 typ.	ASTM D1218

Properties	M –40°C	M –25°C	M –15°C	method
Colour	yellow	yellow	yellow	visual
pH	8.6 typ.	8.5 typ.	8.2 typ.	ASTM D1287
Freeze Point	- 40°C	- 25°C	-15°C	ASTM D 1177
Specific gravity, 20°C	1.071 typ.	1.056 typ.	1.041 typ.	ASTM D1122

CORROSION PROTECTION

Zitrec MC contains an optimized inhibitor package to ensure maximum and long lasting corrosion protection at both high and low temperature. The inhibitors are based on carboxylate technology, which guarantees a longer lifetime than with traditional products.
Anti-corrosion performance is demonstrated through standard and specific corrosion testing.

ASTM D1384 glassware corrosion tests	Weight loss in mg/coupon[1]					
	Brass	Copper	Solder	Steel	Cast Iron	Aluminium
'Industry' limit (max)	10	10	30	10	10	30
Zitrec MC	0.9	1	0.6	0.2	-0.1	0.1

Note 1 : Weight loss AFTER chemical cleaning. Weight gain is indicated by a - sign.

Zitrec Heat transfer fluids

Dynamic heat transfer corrosion test (2000 W)	Weight loss in mg/coupon[1]	
	Cast Iron	Aluminium
test duration, hrs	48	48
Zitrec M-9 [2]		
hot coupon	1.5	23.3
top coupon	2.4	3.6
Zitrec M-40		
hot coupon	-	2.1
top coupon	-	33.3

1 Weight loss AFTER chemical cleaning. Weight gain is indicated by a - sign.
2 Typical test conditions are 20 vol-% dilution

STORAGE REQUIREMENTS

The product should be stored at ambient temperatures and periods of exposure to temperatures above 35°C should be minimized. As with any antifreeze coolant, the use of galvanized steel is not recommended for pipes or any other part of the storage/mixing installation.
Zitrec MC can be stored for minimum 8 years in unopened containers without any effect on the product quality or performance. It is strongly recommended to use new containers and not recycled ones.

TOXICITY & SAFETY

For detailed Toxicity and Safety Data we refer to the Material Safety Data Sheet.
The transport is not regulated. Labeling as for any MEG based heat transfer fluid is required: X_n: R 22 (Harmful if swallowed) and S 2 (Keep out of reach of children).
This product should not be used to protect the inside of drinking water systems against freezing.

Safety Data Sheet

SECTION 1 PRODUCT AND COMPANY IDENTIFICATION

ZITREC MC

Product Use: Antifreeze/Coolant
Product Number(s): 040262

Company Identification
ARTECO N.V.
Technologiepark-Zwijnaarde 2
B-9052 Gent-Zwijnaarde

Belgium

Transportation Emergency Response
Europe: 0044/(0)18 65 407333
Health Emergency
Europe: 0044/(0)18 65 407333
Poison Control Center: 0032/(0)70 245 245
Product Information
Technical Information: 0032/(0)9 240 7320
FAX number: 0032/(0)9 240 7324

SECTION 2 COMPOSITION/ INFORMATION ON INGREDIENTS

COMPONENTS	EC NUMBER	SYMBOL / RISK PHRASES	AMOUNT
Ethylene glycol	203-473-3	Xn/R22	60 - 100 %weight
Sodium 2-ethylhexanoate	243-283-8	Xn/Repro. Cat. 3/R63	1 - 4.9 %weight

The full text of all R-phrases is shown in Section 16. This product contains a bittering agent.

SECTION 3 HAZARDS IDENTIFICATION

CLASSIFICATION: Xn; R22 |
IMMEDIATE HEALTH EFFECTS
Eye: Not expected to cause prolonged or significant eye irritation.
Skin: Contact with the skin is not expected to be harmful.
Ingestion: May be harmful if swallowed.
Inhalation: Not expected to be harmful if inhaled. Breathing this material at concentrations above the recommended exposure limits may cause central nervous system effects. Central nervous system effects may include headache, dizziness, nausea, vomiting, weakness, loss of coordination, blurred vision, drowsiness, confusion, or disorientation. At extreme exposures, central nervous system effects may include respiratory depression, tremors or convulsions, loss of consciousness, coma or death.

DELAYED OR OTHER HEALTH EFFECTS: Not classified.

ENVIRONMENTAL EFFECTS: Not classified.

SECTION 4 FIRST AID MEASURES

Eye: No specific first aid measures are required. As a precaution, remove contact lenses, if worn, and flush eyes with water.
Skin: No specific first aid measures are required. As a precaution, remove clothing and shoes if contaminated. To remove the material from skin, use soap and water. Discard contaminated clothing and shoes or thoroughly clean before reuse.
Ingestion: If swallowed, get medical attention. Do not induce vomiting. Never give anything by mouth to an unconscious person.
Inhalation: No specific first aid measures are required. If exposed to excessive levels of material in the air, move the exposed person to fresh air. Get medical attention if coughing or respiratory discomfort occurs.

SECTION 5 FIRE FIGHTING MEASURES

FLAMMABLE PROPERTIES:
Flashpoint: (Pensky-Martens Closed Cup) 115 °C (239 °F) (Min)
Autoignition: No Data Available
Flammability (Explosive) Limits (% by volume in air): Lower: No data available Upper: No data available

EXTINGUISHING MEDIA: Use water fog, foam, dry chemical or carbon dioxide (CO_2) to extinguish flames. Dry Chemical, CO_2, AFFF Foam or alcohol resistant foam.

PROTECTION OF FIRE FIGHTERS:
Fire Fighting Instructions: This material will burn although it is not easily ignited. For fires involving this material, do not enter any enclosed or confined fire space without proper protective equipment, including self-contained breathing apparatus.
Combustion Products: Highly dependent on combustion conditions. A complex mixture of airborne solids, liquids, and gases including carbon monoxide, carbon dioxide, and unidentified organic compounds will be evolved when this material undergoes combustion.

SECTION 6 ACCIDENTAL RELEASE MEASURES

Protective Measures: Eliminate all sources of ignition in vicinity of spilled material.
Spill Management: Stop the source of the release if you can do it without risk. Contain release to prevent further contamination of soil, surface water or groundwater. Clean up spill as soon as possible, observing precautions in Exposure Controls/Personal Protection. Use appropriate techniques such as applying non-combustible absorbent materials or pumping. Where feasible and appropriate, remove contaminated soil. Place contaminated materials in disposable containers and dispose of in a manner consistent with applicable regulations.
Reporting: Report spills to local authorities as appropriate or required.

SECTION 7 HANDLING AND STORAGE

Specific Use: Antifreeze/Coolant
Precautionary Measures: Do not taste or swallow. Do not breathe vapor or fumes. Keep out of the reach of children.
General Handling Information: Avoid contaminating soil or releasing this material into sewage and

drainage systems and bodies of water.
Static Hazard: Electrostatic charge may accumulate and create a hazardous condition when handling this material. To minimize this hazard, bonding and grounding may be necessary but may not, by themselves, be sufficient. Review all operations which have the potential of generating and accumulating an electrostatic charge and/or a flammable atmosphere (including tank and container filling, splash filling, tank cleaning, sampling, gauging, switch loading, filtering, mixing, agitation, and vacuum truck operations) and use appropriate mitigating procedures.
Container Warnings: Container is not designed to contain pressure. Do not use pressure to empty container or it may rupture with explosive force. Empty containers retain product residue (solid, liquid, and/or vapor) and can be dangerous. Do not pressurize, cut, weld, braze, solder, drill, grind, or expose such containers to heat, flame, sparks, static electricity, or other sources of ignition. They may explode and cause injury or death. Empty containers should be completely drained, properly closed, and promptly returned to a drum reconditioner or disposed of properly.

SECTION 8 EXPOSURE CONTROLS/PERSONAL PROTECTION

GENERAL CONSIDERATIONS:
Consider the potential hazards of this material (see Section 3), applicable exposure limits, job activities, and other substances in the work place when designing engineering controls and selecting personal protective equipment. If engineering controls or work practices are not adequate to prevent exposure to harmful levels of this material, the personal protective equipment listed below is recommended. The user should read and understand all instructions and limitations supplied with the equipment since protection is usually provided for a limited time or under certain circumstances. Refer to appropriate CEN standards.

ENGINEERING CONTROLS:
Use in a well-ventilated area.

PERSONAL PROTECTIVE EQUIPMENT
Eye/Face Protection: No special eye protection is normally required. Where splashing is possible, wear safety glasses with side shields as a good safety practice.
Skin Protection: No special protective clothing is normally required. Where splashing is possible, select protective clothing depending on operations conducted, physical requirements and other substances in the workplace. Suggested materials for protective gloves include: Natural rubber, Nitrile Rubber, Polyvinyl Chloride (PVC or Vinyl).
Respiratory Protection: No respiratory protection is normally required.

Occupational Exposure Limits:

Component	Country/ Agency	TWA	STEL	Ceiling	Notation
Ethylene glycol	EU-Indicative	52 mg/m3	104 mg/m3	--	Skin
Ethylene glycol	United Kingdom	52 mg/m3	104 mg/m3	--	Skin

SECTION 9 PHYSICAL AND CHEMICAL PROPERTIES

Attention: the data below are typical values and do not constitute a specification.

Color: Yellow
Physical State: No Data Available
Odor: Faint or Mild
pH: No data available
Vapor Pressure: No data available

Vapor Density (Air = 1): No data available
Boiling Point: No Data Available
Solubility: **Freezing Point:**
Density: 1.1 kg/l @ 20°C (68°F)
Viscosity: No data available

SECTION 10 STABILITY AND REACTIVITY

Chemical Stability: This material is considered stable under normal ambient and anticipated storage and handling conditions of temperature and pressure.
Incompatibility With Other Materials: May react with strong acids or strong oxidizing agents, such as chlorates, nitrates, peroxides, etc.
Hazardous Decomposition Products: Aldehydes (Elevated temperatures), Ketones (Elevated temperatures)
Hazardous Polymerization: Hazardous polymerization will not occur.

SECTION 11 TOXICOLOGICAL INFORMATION

IMMEDIATE HEALTH EFFECTS
Eye Irritation: The eye irritation hazard is based on evaluation of data for similar materials or product components.

Skin Irritation: The skin irritation hazard is based on evaluation of data for similar materials or product components.

Skin Sensitization: The skin sensitization hazard is based on evaluation of data for similar materials or product components.

Acute Dermal Toxicity: The acute dermal toxicity hazard is based on evaluation of data for similar materials or product components.

Acute Oral Toxicity: The acute oral toxicity hazard is based on evaluation of data for similar materials or product components.

Acute Inhalation Toxicity: The acute inhalation toxicity hazard is based on evaluation of data for similar materials or product components.

ADDITIONAL TOXICOLOGY INFORMATION:
This product contains ethylene glycol (EG). The toxicity of EG via inhalation or skin contact is expected to be slight at room temperature. The estimated oral lethal dose is about 100 cc (3.3 oz.) for an adult human. Ethylene glycol is oxidized to oxalic acid which results in the deposition of calcium oxalate crystals mainly in the brain and kidneys. Early signs and symptoms of EG poisoning may resemble those of alcohol intoxication. Later, the victim may experience nausea, vomiting, weakness, abdominal and muscle pain, difficulty in breathing and decreased urine output. When EG was heated above the boiling point of water, vapors formed which reportedly caused unconsciousness, increased lymphocyte count, and a rapid, jerky movement of the eyes in persons chronically exposed. When EG was administered orally to pregnant rats and mice, there was an increase in fetal deaths and birth defects. Some of these effects occurred at doses that had no toxic effects on the mothers. We are not aware of any reports that EG causes reproductive toxicity in human beings.

2-Ethylhexanoic acid (2-EXA) caused an increase in liver size and enzyme levels when repeatedly administered to rats via the diet. When administered to pregnant rats by gavage or in drinking water, 2-EXA caused teratogenicity (birth defects) and delayed postnatal development of the pups. Additionally, 2-EXA impaired female fertility in rats. Birth defects were seen in the offspring of mice who were administered sodium 2-ethylhexanoate via intraperitoneal injection during pregnancy.

SECTION 12 ECOLOGICAL INFORMATION

ECOTOXICITY
This material is not expected to be harmful to aquatic organisms. The product has not been tested. The statement has been derived from the properties of the individual components.

MOBILITY
No data available.

PERSISTENCE AND DEGRADABILITY
This material is expected to be readily biodegradable. The product has not been tested. The statement has been derived from the properties of the individual components.

POTENTIAL TO BIOACCUMULATE
Bioconcentration Factor: No data available.
Octanol/Water Partition Coefficient: No Data Available

SECTION 13 DISPOSAL CONSIDERATIONS

Use material for its intended purpose or recycle if possible. Oil collection services are available for used oil recycling or disposal. Place contaminated materials in containers and dispose of in a manner consistent with applicable regulations. Contact your sales representative or local environmental or health authorities for approved disposal or recycling methods.
In accordance with European Waste Catalogue (E.W.C.) the codification is the following: 16 01 14

SECTION 14 TRANSPORT INFORMATION

The description shown may not apply to all shipping situations. Consult appropriate Dangerous Goods Regulations for additional description requirements (e.g., technical name) and mode-specific or quantity-specific shipping requirements.

ADR/RID Shipping Description: NOT REGULATED AS DANGEROUS GOODS FOR TRANSPORT UNDER ADR

ICAO/IATA Shipping Description: NOT REGULATED AS DANGEROUS GOODS FOR TRANSPORT UNDER ICAO

IMO/IMDG Shipping Description: NOT REGULATED AS DANGEROUS GOODS FOR TRANSPORT UNDER THE IMDG CODE

SECTION 15 REGULATORY INFORMATION

REGULATORY LISTS SEARCHED:
01=EU. Directive 76/769/EEC: Restrictions on the marketing and use of certain dangerous substances.
02=EU Directive 90/394/EEC: Carcinogens at work.
03=EU Directive 92/85/EEC: Pregnant or breastfeeding workers.
04=EU Directive 96/82/EC (Seveso II): Article 9.
05=EU Directive 96/82/EC (Seveso II): Articles 6 and 7.
06=EU Directive 98/24/EC: Chemical agents at work.

The following components of this material are found on the regulatory lists indicated.

Ethylene glycol	06

CHEMICAL INVENTORIES:
All components comply with the following chemical inventory requirements: AICS (Australia), DSL (Canada), EINECS (European Union), ENCS (Japan), IECSC (China), KECI (Korea), PICCS (Philippines), TSCA (United States).

COUNTRY REGISTRATION:
Denmark: Yes
Italy: Yes
Norway: Yes (P-68550)
Sweden: Yes

CLASSIFICATION - LABELING:
Under the criteria of the directive EEC/67/548 (dangerous substances) and EEC/1999/45 (dangerous preparations):

- contains: Ethylene glycol

Symbols: Xn - Harmful

R22; Harmful if swallowed.
S2; Keep out of the reach of children.
S46; If swallowed, seek medical advice immediately and show this container or label.

SECTION 16 OTHER INFORMATION

REVISION STATEMENT: This is a new Material Safety Data Sheet.
Revision Date: AUGUST 30, 2006

Full text of R-phrases:
R22; Harmful if swallowed.
R63; Possible risk of harm to the unborn child.

ABBREVIATIONS THAT MAY HAVE BEEN USED IN THIS DOCUMENT:

TLV - Threshold Limit Value	TWA - Time Weighted Average
STEL - Short-term Exposure Limit	PEL - Permissible Exposure Limit
CVX - Chevron	CAS - Chemical Abstract Service Number

Prepared according to the criteria of the directive 2001/58/EC by the Chevron Energy Technology Company, 100 Chevron Way, Richmond, California 94802.

The above information is based on the data of which we are aware and is believed to be correct as of the date hereof. Since this information may be applied under conditions beyond our control and with which we may be unfamiliar and since data made available subsequent to the date hereof may suggest modifications of the information, we do not assume any responsibility for the results of its use. This information is furnished upon condition that the person receiving it shall make his own determination of the suitability of the material for his particular purpose.

Technical data

Technical data			NC 630 B
Nominal suction capacity	50 Hz 60 Hz	m^3/h m^3/h	525 630
Ultimate pressure	50 Hz 60 Hz	mbar	≤ 0,05 ≤ 0,01
Nominal motor rating with dis-charge counter pressure between 0,10 and 0,20 bar	50 Hz 60 Hz	kW kW	15,0 17,0
Nominal motor speed	50 Hz 60 Hz	min^{-1} min^{-1}	3000 3600
Noise level (EN ISO 2151)	50 Hz 60 Hz	dB(A) dB(A)	< 70 < 75
Ambient temperature		°C	5 - 50
Maximal counter pressure at the discharge side		bar	0,2
Cooling water requirement		l/ min	12
Cooling water pressure		bar	1 - 6
Cooling water temperature		°C	10 - 30
Weight	env.	kg	550

EC Declaration of Conformity

NOTE: This Declaration of Conformity and the CE-mark affixed to the nameplate are valid for the vacuum pump within the Busch-scope of delivery. When this vacuum pump is integrated into a larger machinery the manufacturer of the larger machinery (this can be operator, too) must conduct the conformity assessment process for the larger machine, issue the Declaration of Conformity for it and affix the CE-mark.

We

Ateliers Busch S.A.
Zone Industrielle
2906 Chevenez
Switzerland

represented in the European Union by

Dr.-Ing. K. Busch GmbH
Schauinslandstr. 1
79689 Maulburg
Germany

declare that the screw vacuum pumps **NC 630 B**

in accordance with the European Directives

“Machinery” 2006/42/EC,

“Electrical Equipment Designed for Use within Certain Voltage Limits” (so called “Low Voltage”) 2006/95/EC,

“Electromagnetic Compatibility” 2004/108/EC

have been designed and manufactured to the following specifications:

Standards	Title of the standard
Harmonised standards	
EN ISO 12100-1 EN ISO 12100-2	Safety of machinery - Basic concepts, general principles of design - Part 1 and 2
EN ISO 13857	Safety of machinery - Safety distances to prevent hazard zones being reached by upper and lower limbs
EN 1012-1 EN 1012-2	Compressors and vacuum pumps - Safety requirements - Part 1 and 2
EN 60204-1	Electrical equipment of machines - Part 1:
EN 61000-6-1 EN 61000-6-3	Electromagnetic compatibility (EMC) – Generic standards – Immunity and emission for residential, commercial and light-industrial environments; Part 1 and 3
EN 61000-6-2 EN 61000-6-4	Electromagnetic compatibility (EMC) – Generic standards – Immunity and emission standard for industrial environments; Part 2 and 4
National standards	
EN ISO 2151	Acoustics - Noise test code for compressors and vacuum pumps - Engineering method (grade 2)

Manufacturer	Mandatory within the EC:
Christian Hoffmann General director	**Dr.-Ing. Karl Busch** General director

Note

Note

Busch – All over the World in Industry

www.busch-vacuum.com

Argentine
Busch Argentina S.R.L.
Santo Domingo 3076
C1293AGN-Capital Federal
Buenos Aires
Tel: +54 11 4302 8183
Fax: +54 11 4301 0896
e-mail: info@busch-vacuum.com.ar

Australia
Busch Australia Pty. Ltd.
30 Lakeside Drive
Broadmeadows, Vic. 3047
Tel: +61 3 93 55 06 00
Fax: +61 3 93 55 06 99
e-mail: sales@busch.com.au

Austria
Busch Austria GmbH
Industriepark Nord
2100 Korneuburg
Tel: +43 2262 / 756 65-0
Fax: +43 2262 / 756 65-20
e-mail: busch@busch.at

Belgium
Busch N.V.
Kruinstraat 7
9160 Lokeren
Tel: +32 9 / 348 47 22
Fax: +32 9 / 348 65 35
e-mail: info@busch.be

Brazil
Busch do Brasil Ltda.
Rod. Edgard Máximo Zambotto, Km 64
13240-000 Jarinu-SP
Tel: +55 11-4016 1400/5277
Fax: +55 11-4016 5399
e-mail: vendas@buschdobrasil.com.br

Canada
Busch Vacuum Technics Inc.
1740, Lionel Bertrand
Boisbriand, Québec J7H 1N7
Tel: +1 450 435 6899
Fax: +1 450 430 5132
e-mail: info@busch.ca

Chile
Busch Chile S. A.
Calle El Roble N° 375-G
Lampa - Santiago
Tel: +56 2 3765136
Fax: +56 2 7387092
e-mail: info@busch.cl

China
Busch Vacuum (Shanghai) Co., Ltd
No.5, Lane 195 Xipu Road
Songjiang Industrial Estate East New Zone
Shanghai 201611 PRC
Tel: +86 (0)21 67600800
Fax: +86 (0)21 67600700
e-mail: busch@busch-china.com

Czech Republic
Busch Vakuum s.r.o.
Pražákova 10
619 00 Brno-Horní Heršpice
Tel: +420 543 42 48 55
Fax: +420 543 42 48 56
e-mail: info@buschpumps.cz

Denmark
Busch Vakuumteknik A/S
Parallelvej 11
8680 Ry
Tel: +45 87 88 07 77
Fax: +45 87 88 07 88
e-mail: info@busch.dk

Finland
Busch Vakuumteknik Oy
Sinikellontie 4
01300 Vantaa
Tel: +358 9 774 60 60
Fax: +358 9 774 60 666
e-mail: info@busch.fi

France
Busch France S.A.S.
16, Rue du Bois Chaland
91090 Lisses
Tel: +33 16989 8989
Fax: +33 16989 8958
e-mail: busch@busch.fr

Germany
Dr.-Ing. K. Busch GmbH
Schauinslandstr. 1
79689 Maulburg
Tel: +49 76 22 6 81-0
Fax: +49 76 22 6 81-194
e-mail: info@busch.de

Dr.-Ing. K. Busch GmbH
Niederlassung Nord
Ernst-Abbe-Str. 1-3
25451 Quickborn
Tel: +49 41 06 7 99 67-0
Fax: +49 41 06 7 99 67-77

Dr.-Ing. K. Busch GmbH
Niederlassung West
Nordring 35
64807 Dieburg
Tel: +49 60 71 92 82-0
Fax: +49 60 71 14 71

Dr.-Ing. K. Busch GmbH
Außenstelle Neuenrade
Breslauer Str. 36
58809 Neuenrade
Tel: +49 23 92 50 29 92
Fax: +49 23 92 50 72 11

Dr.-Ing. K. Busch GmbH
Niederlassung Süd-Ost
Gewerbestraße 3
90579 Langenzenn
Tel: +49 91 01 90 25-0
Fax: +49 91 01 90 25-25

Dr.-Ing. K. Busch GmbH
Außenstelle Zella-Mehlis
Am Rain 11
98544 Zella-Mehlis
Tel: +49 36 82 46 92 71
Fax: +49 36 82 46 92 73

Dr.-Ing. K. Busch GmbH
Außenstelle Meitingen-Ostendorf
Grüntenweg 8
86405 Meitingen-Ostendorf
Tel: +49 82 71 426-341
Fax: +49 82 71 426-342

Hungary
Busch Vacuum Kft.
Bentonit u. 8
1225 Budapest
Tel: +36 1 207 6135
Fax: +36 1 207 6136
e-mail: busch@busch-vacuum.hu

India
Busch Vacuum India Pvt Ltd.
Plot No. 110, Sector 7
PCNTDA, Bhosari
Pune 411026
Tel: +91 206410 2886
Fax: +91 202711 2838
e-mail: sales@buschindia.com

Ireland
Busch Ireland Ltd.
A10-11 Howth Junction Business Centre
Kilbarrack, Dublin 5
Tel: +353 1 832 1466
Fax: +353 1 832 1470
e-mail: sales@busch.ie

Israel
Busch Israel Ltd.
1 Mevo Sivan Street
Qiryat Gat 82022, Israel
Tel: +972 (0)8 6810485
Fax +972 (0)8 6810486
e-mail: service_sales@busch.co.il

Italy
Busch Italia S.r.l.
Via Ettore Majorana, 16
20054 Nova Milanese
Tel: +39 0362 370 91
Fax: +39 0362 370 999
e-mail: info@busch.it

Japan
Nippon Busch K.K.
1-23-33, Megumigaoka
Hiratsuka City, Kanagawa
Japan 259-1220
Tel: +81 463-50-4000
Fax: +81 463-50-4004
e-mail: info@busch.co.jp

Korea
Busch Korea Ltd.
248-2, Ichi-ri, Majang-Myun,
Icheon-si, Kyunggi-Do
Tel: +82 31 321 8114
Fax: +82 31 321 8877
e-mail: busch@buschkorea.co.kr

Malaysia
Busch Malaysia Sdn Bhd.
4&6, Jalan Taboh 33/22, Seksyen 33
Shah Alam Technology Park
40400 Shah Alam
Selangor Darul Ehsan
Tel: +60 3 5122 2128
Fax +60 3 5122 2108
e-mail: busch@busch.com.my

Mexico
Busch Vacuum Mexico S. de R.L. de C.V.
Tlaquepaque 4865, Los Altos
Monterrey, Nuevo Leon
Mexico 64370
Tel: +52 81 8311-1385
Fax: +52 81 8311-1386
e-mail: info@busch.com.mx

Netherlands
Busch B.V.
Pompmolenlaan 2
3447 GK Woerden
Tel: +31 348-462300
Fax: +31 348-422939
e-mail: info@busch.nl

New Zealand
Busch New Zealand Ltd.
Unit D, 41 Arrenway Drive
Albany, Auckland 1330
Tel: +64 9 414 7782
Fax: +64 9 414 7783
e-mail: sales@busch.co.nz

Norway
Busch Vakuumteknikk AS
Hestehagen 2
1440 Drøbak
Tel: +47 64 98 98 50
Fax: +47 64 93 66 21
e-mail: busch@busch.no

Poland
Busch Polska Sp. z o.o.
Ul. Chopina 27
87-800 W»oc»awek
Tel: +48 54 2315400
Fax: +48 54 2327076
e-mail: busch@busch.com.pl

Portugal
Busch Ibérica S.A., Sucursal em Portugal
Zona Industrial Norte, Fracção B, Armazém 2
3750-753 Raso de Travassô - Agueda
Aveiro, Portugal
Tel: +351 234 648 070
Fax: +351 234 648 068
e-mail: geral@buschib.pt

Russia
Busch Vacuum Russia OOO
Kotlyakovskaya str., 6/9
115201 Moscow
Tel: +7 495 6486726
Fax: +7 495 6486724
e-mail: info@busch.ru

Singapore
Busch Vacuum Singapore Pte Ltd
20 Shaw Road
Unit 01-03 Ching Shine Building
Singapore 367956
Tel: +65 6488 0866
Fax: +65 6288 0877
e-mail: busch@busch.com.sg

South Africa
Busch Vacuum South Africa (Pty) Ltd.
Denver
Johannesburg
Tel: +27 11 856 0650/6
Fax: +27 11 856 0625
e-mail: joe.jagger@busch.co.za

Spain
Busch Ibérica S.A.
Pol. Ind. Coll de la Manya
C/ Jaume Ferran, 6-8
08403 Granollers
Tel: +34 93 861 61 60
Fax: +34 93 840 91 56
e-mail: busch@buschib.es

Sweden
Busch Vakuumteknik AB
Bråta Industriområde
435 33 Mölnlycke
Tel: +46 31-338 00 80
Fax: +46 31-338 00 89
e-mail: info@busch.se

Switzerland
Busch AG
Waldweg 22
4312 Magden
Tel: +41 61 / 845 90 90
Fax: +41 61 / 845 90 99
e-mail: info@buschag.ch

Taiwan
Busch Taiwan Corporation
1F. No. 69, Sec. 3, Beishen Road
Shenkeng Township,
Taipei County 222
Tel: +886 2 2662 0775
Fax: +886 2 2662 0796
e-mail: info@busch.com.tw

Thailand
Busch Vacuum (Thailand) Co., Ltd.
888/30 Moo19, Soi Yingcharoen,
Bangplee-Tamru Road,
Bangpleeyai, Bangplee,
Samutprakarn 10540
Tel: +66 2-382-5428
Fax: +66 2-382-5429
e-mail: info@busch.co.th

Turkey
VAKUTEK
Emlak Kredi Ishani No: 179
34672 Üsküdar-Istanbul
Tel: +90 216 310 0573
Fax: +90 216 343 5126
e-mail: vakutek@ttnet.net.tr

United Kingdom
Busch (UK) Ltd
Hortonwood 30
Telford Shropshire TF1 7YB
Tel: +44 1952 677 432
Fax: +44 1952 677 423
e-mail: sales@busch.co.uk

USA
Busch LLC
516-B Viking Drive
Virginia Beach, VA 23452
Tel: +1 757 463-7800
Fax: +1 757 463 7407
e-mail: marketing@buschusa.com